

The Next Industrial Age

The Future of Industrial Design in
Western Legacy Industries

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To the engineers, designers, and makers
who keep the old machines running
while building the ones that will replace them.

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Foreword: The Steel That Still Holds

THE FACTORY HAS BEEN HERE FOR SIXTY YEARS. THE MACHINES INSIDE IT ARE OLDER THAN MOST OF THE PEOPLE WHO OPERATE THEM. THE COMPANY THAT BUILT THEM HAS CHANGED HANDS THREE TIMES. AND YET, ON A GREY TUESDAY MORNING, IT IS STILL PRODUCING — STILL CASTING, FORGING, MACHINING, ASSEMBLING. STILL EMPLOYING PEOPLE. STILL PAYING TAXES. STILL PART OF THE FABRIC OF THE TOWN AROUND IT.

This is the reality of Western legacy industry. It is not the gleaming vision of Industry 4.0 that fills conference brochures. It is older, slower, and far more complicated. And it is precisely this complexity that makes it the most important design challenge of our time.

For decades, the story of industrial design has been told through consumer products — the smartphone, the electric car, the smart speaker. Meanwhile, the industries that actually underpin Western economies — aerospace, heavy machinery, chemicals, energy, shipbuilding, defence — have been treated as engineering problems rather than design opportunities.

This book argues that the opposite is true. The next industrial age will not be won by technology alone. It will be won by how we design the integration of technology into industries that were built for a different era. It will be

won by designing for the workers who run these industries, the communities that depend on them, and the environmental realities that now constrain them.

This is a book about the future of making things in the West. It is written for designers, engineers, executives, policymakers, and anyone who believes that the industrial age is not over — only beginning again.

Toolkit Summary

- Legacy industries are not dying; they are transitioning. Design will determine how well.
- Industrial design must expand beyond consumer products to the systems that underpin Western economies.
- The next industrial age will be won through integration, not invention alone.

Chapter 1: Legacy Industries at a Crossroads

WALK ONTO THE SHOP FLOOR OF A WESTERN HEAVY ENGINEERING PLANT AND YOU WILL SEE SOMETHING THAT CONTRADICTS EVERY HEADLINE YOU HAVE READ FOR THE PAST DECADE. NOT RUST AND ABANDONMENT — PRODUCTION. MACHINES STAMPING, WELDING, WINDING, AND ASSEMBLING. PALLETS MOVING. FLUORESCENT LIGHT OVER THREE SHIFTS. THE ECONOMY'S NARRATIVE ABOUT THE WEST HAS BEEN ONE OF DEINDUSTRIALISATION, BUT THE REALITY IS MORE PRECISE: THE WEST HAS NOT STOPPED MAKING THINGS. IT HAS STOPPED MAKING NEW THINGS, AND IT HAS STOPPED REDESIGNING THE THINGS IT MAKES.

This is the defining condition of legacy industry. The term covers the sectors that built the modern West and still carry its weight: aerospace, automotive, heavy machinery, chemicals, energy, shipbuilding, and defence. These industries share a set of traits that make them a single problem, even though they occupy different corners of the economy. They are capital-intensive, with assets measured in decades rather than years. They are highly regulated, because failure is not a bug in the software but a ship, a refinery, or a nuclear plant. They employ millions of people in roles that cannot be

offshored to a call centre. And they are, almost without exception, running on infrastructure and knowledge that was designed in the twentieth century.

None of this makes them obsolete. It makes them neglected. And neglect, in an industrial context, is not a passive state — it is an active erosion of competitiveness, capability, and confidence.

The State of Play

Consider the assets first. The average age of the industrial base in most Western economies is striking. In the United States, manufacturing capacity utilisation fluctuates around levels that would have been considered a recession in the 1970s, and much of the physical plant dates from an era when a different set of assumptions governed energy prices, labour costs, and environmental regulation. European industry is older still in places — the Ruhr, the Midlands, northern Italy, the French and Spanish shipbuilding coasts — where factories and yards built in the postwar boom are still producing, often with control systems that predate the personal computer.

Ageing assets are not inherently a problem. Some of the best manufacturing in the world happens in old buildings with new thinking inside. But ageing assets combined with ageing skills is a different matter. The workforce that built and ran these industries is retiring. In aerospace, the engineering population skews heavily toward employees with decades of experience — precisely

the cohort now leaving. In nuclear energy, utilities in Europe and North America have spent years begging for the engineers needed even to decommission old plants, let alone build new ones. The UK's nuclear regulator has warned openly about the "critical skills gap" in the sector. The German Mittelstand — the family-owned engineering firms that are the backbone of European manufacturing — reports that finding qualified engineers and skilled machinists is now its single biggest constraint on growth, ahead of energy costs and interest rates.

Meanwhile, the competitive pressure from Asia has changed character. It is no longer about cheap labour alone. Chinese shipbuilding now dominates global order books; Korean yards lead in the most complex vessels; Japanese and Chinese firms have moved up the value chain in machine tools, chemicals, and increasingly aerospace. What the West once dismissed as low-cost imitation is now world-class engineering at scale, backed by state-directed capital and a domestic market large enough to fund iteration. The competition is not unfair in any simple sense. It is simply a different model — and it has been winning.

Why These Industries Matter Disproportionately

It is fashionable in some policy circles to treat legacy industry as a managed decline problem — something to be wound down gently while the knowledge economy

takes over. This view is not merely wrong; it is dangerous, because it ignores three ways these industries matter out of all proportion to their share of GDP.

The first is employment. Manufacturing still supports a vastly larger share of jobs than its headline GDP number suggests, because each direct industrial job supports a chain of suppliers, logistics, maintenance, and local services. In Germany, roughly one in four jobs is tied to industry. In the United States, manufacturing employs around 13 million people directly — and those jobs pay, on average, significantly more than the service-sector alternatives available to the same workers. These are the jobs that anchor the middle class in places that have no financial district and no tech campus. When a shipyard closes, it does not close a factory; it closes a town's future.

The second is national security. Aerospace, defence, shipbuilding, energy, and advanced materials are not ordinary tradable goods. They are the industrial base on which military capability rests — what strategists call the defence industrial base. A country that cannot build ships, engines, turbines, or precision components domestically cannot sustain its armed forces in a conflict of any duration. The war in Ukraine made this brutally concrete: European defence ministries discovered that their munitions stockpiles were thin precisely because their industrial bases had been allowed to shrink. Restocking them required reopening production lines that had been mothballed for years, and rebuilding workforces that had

dispersed. The lesson is that industrial capacity is a strategic asset that decays when it is not used — and it cannot be bought back quickly at any price.

The third is export value. Legacy industries remain the West's biggest earners in global trade. Aerospace is the largest net export sector of the United States. Germany's exports are dominated by vehicles, machinery, and chemicals. France's by aerospace and luxury goods with an industrial core. These are the sectors that pay for imports of energy, consumer electronics, and the other goods the West no longer makes. A decline in these exports is not a statistical curiosity; it is a slow-motion transfer of national wealth.

The Three Futures

Every legacy industry in the West is now facing the same fork in the road, and the three possible futures are distinct enough to name.

The first future is **decline**. This is the path of least resistance: continue to underinvest, fail to replace retiring expertise, lose order after order to Asian competitors, and shrink. It rarely announces itself as a decision. It happens through a thousand small deferrals — the maintenance postponed, the apprentice intake cut, the R&D budget trimmed "temporarily," the design department merged into engineering and then outsourced. Decline is not dramatic. It is a compound-interest process running in reverse. Some industries have

already travelled most of the way down this path — much of Western consumer electronics, most of Western shipbuilding in the 1980s and 1990s, and, more recently, parts of European solar and battery manufacturing. The pattern is always the same: capability is lost first, then capacity, then the knowledge of how to rebuild either.

The second future is **stagnation**. This looks healthier than decline, because the headline numbers hold. Output is stable. Employment is stable. The industry remains profitable — often very profitable — by defending niches, protected markets, and existing customer relationships. But stagnation means serving a shrinking definition of the market. It means building the same products with the same processes, slightly improved, and watching the frontier of what is possible move elsewhere. Stagnant industries are comfortable for the executives who run them and miserable for the young engineers who join them, because there is nothing to design. The cost of stagnation is deferred, but it is not cancelled: it shows up a decade later as an industry that has quietly lost the ability to compete for anything except the work it already has.

The third future is **reinvention**. This is the thesis of this book: legacy industries, with their assets, skills, and market positions, are not prisoners of their history. They are platforms. The same heavy engineering capability that built twentieth-century industry can build the next generation of energy systems, defence capability, transport, and manufacturing itself — if it is redesigned

rather than merely maintained. Reinvention is not about abandoning the old; it is about rebuilding on top of it. It is the difference between replacing a machine and reimagining what the machine is for. This book is about how design — the discipline of deciding what to build and why — can make the third future more than a slogan.

The Handover Problem

Before any of that can happen, legacy industries have to survive the most underrated crisis they face: the transfer of knowledge from one generation to the next. This is not a human-resources problem. It is a systems problem, and it is happening on a scale the West has never managed before.

The mechanics are unforgiving. In aerospace, a significant share of the most experienced engineers and skilled machinists are at or past retirement age. In nuclear, the situation is acute: the people who built the current fleet are leaving, and the people who would replace them were not hired during the decades when the sector was in decline. In heavy machinery and chemicals, the pattern is identical — a compressed cohort of experts in their sixties, a missing generation in their forties and fifties, and a thin layer of newcomers in their twenties and thirties.

The missing generation matters more than the retiring one. Expertise in these industries is not book knowledge. It is accumulated judgment: the machinist who knows

how a particular fifty-year-old lathe behaves at the edge of its tolerances, the engineer who remembers why a design decision was made thirty years ago and what would break if it were changed, the metallurgist who can read a casting's history in its surface. This knowledge lives in heads, not databases. When it retires, it does not go to a competitor — it simply ceases to exist.

The handover is failing for three reasons. First, the knowledge was never written down, because there was never time and never a perceived need. Second, the companies that hold it have spent decades cutting training budgets, so the mechanisms for transfer — apprenticeships, mentoring, structured progression — are skeletal. Third, the young people who do enter these industries are often steered toward the newest technology and away from the installed base, which means the legacy knowledge is not just unrecorded but actively devalued.

This is where the design mindset changes the terms of the problem. A handover is a design problem: it requires deliberately designing the systems, interfaces, documentation, and training through which expertise moves from one generation to the next. It requires designing machines that are legible to newcomers — that explain themselves, monitor themselves, and capture the knowledge of their operators. It requires treating the retiring generation not as a cost to be retired but as an asset to be mined. The companies that solve the handover

will not merely survive; they will have a durable advantage, because their competitors' expertise is walking out the door.

The Crossroads Is Now

None of this is hypothetical. The three futures are not distant scenarios; they are already differentiating. Some Western shipbuilders and their suppliers are winning complex, high-value work; others are shrinking to repair and niche contracts. Some chemical companies are redesigning their plants and portfolios for a decarbonised world; others are defending the old model into obsolescence. Some aerospace firms are investing in the next generation of propulsion and manufacturing; others are managing their legacy products into the sunset.

The difference between these outcomes is not mainly technology, capital, or policy — though all matter. The difference is whether anyone is asking the design question: what is this industry for, and what should it become? Legacy industries at the crossroads will not be saved by more of the same. They will be saved by the willingness to redesign themselves — their products, their processes, their relationship with workers, and their place in the world.

That is the argument of this book. The chapters ahead take it apart piece by piece: the new role of industrial design, the digital-physical frontier, sustainability as a design constraint, the materials revolution, and the

human dimension of automation. The conclusion is consistent throughout. The industrial age is not ending in the West. It is being redesigned — and the design decisions are being made right now, on shop floors and in boardrooms, by people who may not think of themselves as designers at all.

The factory has been there for sixty years. The question is what it becomes in the next sixty. That question will be answered by design, whether or not anyone in the room is called a designer.

Toolkit Summary

- Legacy industries — aerospace, automotive, heavy machinery, chemicals, energy, shipbuilding, defence — share a common condition: ageing assets, retiring expertise, and intensifying Asian competition.
- They matter disproportionately: they anchor middle-class employment, underpin national security, and generate the bulk of Western export value.
- The three futures are decline (compound erosion), stagnation (comfortable irrelevance), and reinvention (rebuilding on the existing base) — and all three are already underway in different sectors.

- The cross-generational handover is the most urgent and most neglected problem: expertise lives in retiring heads, not databases, and the transfer mechanisms have been allowed to atrophy.
- Treating the handover as a design problem — legible machines, captured knowledge, deliberately engineered training — turns a looming crisis into a durable competitive advantage.
- The crossroads is not in the future; the decisions that determine which future each industry gets are being made now.

Chapter 2: The New Role of Industrial Design

FOR MOST OF THE TWENTIETH CENTURY, "INDUSTRIAL DESIGN" MEANT ONE THING: THE STYLING OF MANUFACTURED OBJECTS. A DESIGNER WAS THE PERSON WHO MADE THE PRODUCT LOOK GOOD — THE CURVE OF A CAR BODY, THE SWEEP OF A DASHBOARD, THE COLOURWAY OF A VACUUM CLEANER. THE DISCIPLINE SAT AT THE SURFACE OF INDUSTRY, ADDING VALUE AT THE POINT OF SALE AND LITTLE ANYWHERE ELSE. ENGINEERS REGARDED DESIGNERS AS DECORATORS; EXECUTIVES REGARDED THEM AS A COST LINE TO BE MINIMISED; AND THE INDUSTRIES THAT DIDN'T SELL DIRECTLY TO CONSUMERS — CHEMICALS, ENERGY, HEAVY MACHINERY, DEFENCE — REGARDED THEM AS IRRELEVANT ALTOGETHER.

That division of labour is collapsing, and its collapse is one of the most important shifts in the industrial landscape. The design challenge of the next industrial age is not how a product looks. It is how an entire industrial system — a plant, a supply chain, a product's full life cycle, the ecosystem of machines and people that operates it — is conceived, integrated, and operated. The companies that understand this are already pulling away from those that don't. The ones that don't are still hiring people to make their control panels prettier.

From Styling Products to Designing Systems

The shift begins with a change in the object of design. A product — a turbine, a compressor, an excavator — is a bounded thing. You can hold it, ship it, sell it. A system is not. A system is the product plus everything that touches it: the sensors that monitor it, the software that schedules its maintenance, the operators who run it, the grid it connects to, the regulations that govern it, the data it generates, and the second life it will have when its primary use ends.

Designing systems is a different discipline from styling products. It requires thinking about flows rather than objects — flows of material, energy, information, and money. It requires designing for time: a product is designed for the moment of sale; a system is designed for thirty years of operation, evolution, and eventual decommissioning. And it requires designing for many stakeholders at once — the operator on the night shift, the maintenance crew, the regulator, the customer's customer, the community around the plant — where a product only needed to satisfy the buyer.

The most instructive examples come from the energy transition. An offshore wind turbine is, at one level, a piece of heavy machinery — a gearbox, blades, a tower, a foundation. But the companies that win in offshore wind are not the ones with the best gearbox; they are the ones that design the whole system: the turbine, the array

layout, the installation vessel logistics, the operations-and-maintenance strategy, the grid connection, and the decommissioning plan, as a single integrated proposition. Vestas and Siemens Energy compete on system design, not component performance. The same logic applies to electric-vehicle battery plants, hydrogen electrolyzers, and every other new industrial asset being built in the West today. The product is the entry ticket. The system is the game.

Design Thinking Meets the Plant Floor

The second shift is methodological. "Design thinking" — the human-centred, iterative, prototype-early approach popularised by IDEO and adopted wholesale by the tech industry — has, until recently, lived in the world of software, services, and consumer products. Its industrial application has been close to zero. That is changing, and the change is overdue, because the problems legacy industry faces are precisely the problems design thinking was built to solve: complex systems with multiple stakeholders, poorly articulated user needs, and failure modes that only show up in use.

Take the operator interface of a chemical plant. For decades, these were designed by control-systems engineers who thought in terms of process variables, alarm limits, and screen real estate. The result, universally, is a control room with thousands of alarms — so many that operators learn to ignore them, and

genuinely critical alerts get lost in the noise. This is not an engineering failure; it is a design failure. It is the direct result of nobody asking the user-centred question: what does an operator actually need to know, at 3 a.m., with a plant running at 90 percent load, to make the right decision? Companies like Honeywell and Yokogawa have spent the last decade redesigning alarm systems around operator cognition — prioritising, filtering, and presenting information the way a good mentor would. The safety and performance gains have been real, and they came from applying a design lens to a problem that had been treated as pure engineering.

The same applies to maintenance. Maintenance procedures for heavy equipment are typically written by engineers for engineers, in dense technical prose, and delivered as PDFs. Designers would ask a different question: what does the technician need, in the field, gloved and under time pressure? The answer is increasingly visual, step-by-step, context-aware guidance delivered through the equipment itself or an augmented-reality headset — the approach pioneered by companies like PTC and by aerospace manufacturers who now train and support maintenance crews with AR overlays on complex assemblies. This is design thinking applied to industrial processes, and it is not cosmetic. It is a productivity technology.

From Product-Centric to Ecosystem-Centric

The third shift is the most strategic: the unit of value has moved from the product to the ecosystem around it. The classic formulation of this shift is the "servitisation" of industry — the move from selling a machine to selling the outcome the machine delivers. Rolls-Royce's "Power by the Hour" model, under which airlines pay for engine operating hours rather than buying engines outright, is the canonical example. The manufacturer retains ownership, monitors the engine in flight, schedules maintenance proactively, and is financially responsible for reliability. The product became the entry point to a service relationship.

But the ecosystem shift goes beyond services. It means the product's design must now anticipate its place in a network of other products, platforms, and stakeholders. A construction excavator is designed differently when it is expected to feed its own operational data into a fleet-management platform, coordinate with autonomous haulage on site, and comply with emissions reporting in multiple jurisdictions. A defence platform is designed differently when it must interoperate with allied systems for its entire life. An industrial robot is designed differently when it is expected to work alongside people rather than behind a fence.

Ecosystem-centric design changes what "good design" means at the most fundamental level. In the product-centric era, the design goal was differentiation — making the product visibly, desirably different from the competitor's. In the ecosystem-centric era, the design goal is often the opposite: interoperability, compatibility, and graceful integration. The best-designed product in the new era may be the one that disappears most smoothly into a larger system. This is a hard adjustment for industrial cultures that spent a century competing on the virtues of the standalone machine.

The Designer as Integrator

If the object of design has expanded from product to system, the role of the designer has expanded with it. The industrial designer of the next age is not the person who makes things attractive. They are the integrator — the person who can move between engineering, business, and users, translate between them, and hold the whole picture in view.

This is a genuinely difficult role, because each of those communities speaks a different language and holds a different definition of success. Engineers optimise for performance and manufacturability; their instinct is to solve the technical problem and declare victory. Businesspeople optimise for margin, market position, and risk; their instinct is to protect the existing revenue stream. Users optimise for usability, safety, and dignity at

work; their voice is the quietest in the room and the most rarely invited. The integrator's job is to make sure the solution is not just technically sound and commercially viable, but actually usable — and to make the trade-offs between those goals explicit rather than accidental.

The best industrial designers have always done some of this, but the profession's self-conception has rarely caught up. In legacy industry, the integrator role is more often played by accident — by a grizzled chief engineer who happens to understand customers, or a project manager who happens to care about operators. The opportunity is to make integration a deliberate, staffed, respected function: a design function in the fullest sense.

This is why the design skill set — synthesis, user research, prototyping, visual communication, systems thinking — is the missing ingredient in legacy industries. They have world-class engineering. What they lack is the connective tissue: the people who can take a thirty-year-old product line and a new technology and a retiring workforce and a demanding customer and shape them into a coherent whole.

Why Legacy Industries Need Design Leadership

It is worth being blunt about the current state of design in legacy industry, because the gap is stark. In consumer technology, design is a board-level function. Apple, Tesla, and their peers treat design leadership as a competitive

weapon; design decisions are CEO decisions. In legacy industry, design — where it exists at all — is typically a small department reporting into engineering or marketing, staffed by people whose job description is closer to drafting than to strategy.

The result is the paradox at the heart of this book: the industries that most need design leadership have the least of it. Consumer products operate in mature, crowded markets where design is the difference between winning and losing; they invested accordingly. Legacy industries operate in markets that are suddenly being disrupted — by decarbonisation, by digitalisation, by geopolitical shifts, by a generational change in the workforce — and they have no equivalent capability. A chemical company facing the redesign of its entire product portfolio for a low-carbon world is facing the biggest design problem of its existence, and it will typically have zero senior designers on the payroll.

The consumer-industrial gap is visible in the workforce itself. Young people entering these industries are digital natives with design literacy from consumer products. They experience a jarring mismatch: their phones are beautifully designed, and the machine they are asked to operate was designed before they were born, with interfaces, documentation, and workflows that have barely changed since. This mismatch is a retention problem — young engineers leave industries that feel stuck in the past — and it is a safety problem, because poorly designed interfaces fail under stress. The gap

between consumer design sophistication and industrial design neglect is not an aesthetic issue. It is a competitiveness issue, and it is measurable in lost productivity, lost talent, and lost orders.

Design as the Competitive Frontier

There is a reason the most successful industrial companies of the last two decades have been the ones that took design seriously. Airbus redesigned its entire production system around digital design and assembly integration. Tesla, whatever one thinks of its manufacturing travails, proved that a design-led approach could force the entire automotive industry to rethink the vehicle as a software-defined, battery-driven system. Siemens, GE, and Schneider Electric have repositioned themselves from equipment vendors to design-and-integration partners for the digitalised industrial world.

The pattern is consistent: in every case, the design question — what is this thing for, who uses it, what system does it belong to, and how should it evolve — was asked at the top of the company and answered with investment, not decoration. The next industrial age will reward exactly that behaviour. The companies that treat design as the integration function between engineering, business, and users will design the future. The ones that treat it as styling will be designed by it.

Toolkit Summary

- The object of industrial design has shifted from the product to the system: products plus sensors, software, operators, regulation, data, and second life.
- Design thinking — user-centred, iterative, prototype-early — is directly applicable to industrial processes like control-room interfaces, alarm systems, and maintenance procedures, where poor design costs real money and real safety.
- The unit of value has moved from the standalone machine to the ecosystem: servitisation (Rolls-Royce's Power by the Hour), interoperability, and graceful integration now drive design goals.
- The designer's new role is integrator — translating between engineering, business, and users, and making trade-offs explicit rather than accidental.
- Legacy industries have world-class engineering but minimal design leadership; the consumer-industrial design gap is a measurable competitiveness and retention problem.
- Design-led companies (Airbus, Tesla, Siemens, Schneider) are out-competing engineering-led ones in every disrupted industrial market.

Chapter 3: The Digital-Physical Frontier

EVERY INDUSTRIAL PRODUCT NOW EXISTS TWICE: ONCE AS ATOMS, ONCE AS BITS. THE PHYSICAL MACHINE ON THE SHOP FLOOR HAS A DIGITAL COUNTERPART SOMEWHERE — A 3D MODEL IN A CAD SYSTEM, A RECORD IN AN ERP DATABASE, A STREAM OF SENSOR DATA, A SIMULATION IN A SERVER FARM. THE RELATIONSHIP BETWEEN THOSE TWO VERSIONS OF REALITY IS THE DIGITAL-PHYSICAL FRONTIER, AND IT HAS BECOME THE CENTRAL ARENA OF INDUSTRIAL COMPETITION. THE COMPANIES THAT MASTER IT DO NOT JUST OPERATE THEIR ASSETS MORE EFFICIENTLY; THEY REDESIGN THEM, AND REDESIGN THE BUSINESSES AROUND THEM. THE COMPANIES THAT DON'T MASTER IT ARE FLYING BLIND, WHETHER THEY KNOW IT OR NOT.

The frontier has a history. The first CAD systems of the 1960s and 1970s replaced drawing boards with geometric models — a revolution in documentation, but not in understanding: the digital model was still a static picture of a static design. The 1980s brought CAE — computer-aided engineering — which let engineers simulate stress, fluid flow, and thermal behaviour before anything was built, and CAM, which translated geometry into machine toolpaths. By the 1990s, product data management tied these tools together into a coherent digital record of the product. Each step added capability, but the relationship

between digital and physical remained one-way: the digital model described the product, and then the product was built and the model went stale. Nobody updated the model with what actually happened in manufacturing and in service.

The digital twin is the inversion of that relationship. Instead of a one-way description, the twin is a two-way mirror: a living digital model that is continuously updated with data from the physical asset, that can be used to simulate what the asset will do next, and that can drive decisions back into the physical world. The term was popularised by Michael Grieves in the early 2000s and adopted with missionary zeal by the technology industry, but the underlying idea — that you cannot manage what you cannot model — is as old as engineering. What changed is that the model can now be kept alive.

The Anatomy of a Digital Twin

A serious digital twin has three parts. The first is the geometric and engineering model — the CAD heritage: what the asset is made of, how it is shaped, how it was designed to behave. The second is the live data layer: sensors reporting temperature, vibration, pressure, position, power draw, and a hundred other variables in real time. The third is the simulation and analytics engine that fuses the two — that takes the model, the data, and the operating context, and produces insight: this bearing will fail in six weeks; this blade is operating outside its

design envelope; this compressor is running 3 percent below optimal efficiency and here is the setting that fixes it.

The value of the twin scales with the cost and criticality of the asset. For a consumer product, a twin is a nice-to-have. For a gas turbine, a container ship, a blast furnace, or a military platform, it is transformative, because the cost of downtime, failure, or suboptimal operation is measured in millions per day. This is why the earliest and best digital-twin implementations are in the most expensive assets.

What Works: The Cases

The canonical example remains Rolls-Royce's engine health monitoring. Every modern Rolls-Royce aero engine is festooned with sensors — hundreds of them, measuring temperatures, pressures, vibrations, and speeds throughout the engine cycle. That data is streamed to the ground in flight, analysed against models of engine behaviour built up over decades, and used to decide when an engine needs attention, what it needs, and where the work should be done. The result is that engine failures are predicted, not discovered; maintenance is scheduled around airline operations rather than imposed on them; and the engine maker carries a model of every engine in service, refined by every flight it makes. This is the

"Power by the Hour" model made physically possible: you can only sell operating hours if you know, in detail, how the engine is operating.

The same logic has spread across heavy industry. Siemens, through its MindSphere platform, built its pitch around connecting industrial assets — from wind turbines to machine tools to entire factories — into a common data environment, and using that data to optimise performance and predict failures. GE Digital's Predix was the most ambitious attempt: a purpose-built industrial IoT platform that GE intended to make the operating system of the industrial world, connecting turbines, locomotives, hospitals, and power plants. Predix's commercial fate — it was wound down and GE Digital was divested — is itself instructive. It failed not because the technology was wrong but because the business model was: a giant platform bet requires network effects that no single vendor, even one with GE's asset base, could create on its own, and the company's internal turmoil made the multi-year investment untenable. The lesson for legacy industry is not that digitalisation doesn't work. It is that digitalisation works when it is anchored to specific assets and specific problems, and fails when it is sold as a general-purpose platform awaiting users.

That distinction — anchored versus abstract — is the difference between the successes and failures across the industry. The wind-farm operator that uses blade sensors and turbine twins to predict component wear and schedule maintenance before failure is doing

digitalisation that pays for itself. The conglomerate that builds a data lake and waits for insight to emerge is doing digitalisation that burns money. Digital-physical integration is a design discipline, not a procurement category.

The Role of IoT in Legacy Machinery

The most important development for legacy industry is not the greenfield digital factory, which is rare and expensive. It is the retrofit: taking machinery designed before the internet existed and giving it a digital voice. Industrial IoT has made this cheap and practical. Vibration sensors the size of a coin, wireless gateways, edge computers the size of a paperback, and cloud analytics that cost pennies per asset per day now allow a fifty-year-old machine tool, compressor, or pump to report on its own health.

This is a design decision disguised as a technology decision. Retrofitting sensors is not just instrumentation; it is deciding what the machine should be allowed to say and who should listen. The design questions are practical: Which failure modes are worth predicting? Vibration analysis catches bearing wear and imbalance; oil analysis catches contamination and chemistry drift; current signature analysis catches electrical and load issues; thermography catches insulation and cooling failures. Each sensor type is a bet on which failures matter. The design questions are also organisational: the data is only

valuable if someone is accountable for acting on it. The plants that succeed with predictive maintenance are the ones where the digital system outputs not reports but work orders — where insight is wired directly into the maintenance workflow rather than left in a dashboard. The plants that fail are the ones where the dashboard is admired and ignored.

Predictive maintenance, done properly, changes the economics of legacy assets. Instead of preventive maintenance on a fixed schedule — replacing parts that still have life, at intervals chosen by conservatism — the asset is maintained at the point of need, with the part and the crew already organised. Downtime falls, part life extends, and — critically for the skills question — the tacit knowledge of experienced maintainers gets codified into the model, partly replacing the expertise that is retiring. The machine that can predict its own failure no longer depends entirely on the person who remembers how it failed last time.

The Digital Thread

The digital twin is the live model of an asset in operation. The digital thread is the connective tissue that links every stage of the asset's life — design, engineering, manufacturing, supply chain, operation, maintenance, and end of life — through a continuous, traceable stream

of data. Where the twin is a snapshot of the present, the thread is the complete history, and it matters for reasons that go far beyond efficiency.

The thread makes traceability possible. In aerospace and defence, where every component's provenance, pedigree, and test history must be documented, the digital thread turns a paperwork burden into a data asset. In regulated industries, it turns compliance from an after-the-fact exercise into a by-product of normal operation. And in the emerging circular economy, the thread is the basis for the materials passport — the record of what a product is made of, how it was made, and what it can become — without which recycling and reuse at industrial scale are impossible.

The thread also changes how products are designed, because it closes the feedback loop. For the first time, the designer can see what actually happens to a design over its life: how it is manufactured, where it fails, how it is maintained, how it is eventually disposed of. This is the difference between designing from assumptions and designing from evidence. Every company that closes the loop discovers the same thing: reality diverges from the design intent in ways that were always invisible before. The turbine blade that fails at a specific operating point. The seal that degrades faster in a particular climate. The component that is never replaced because it is too hard to reach. Design decisions made on the basis of the digital thread are simply better than design decisions made on the basis of anything else.

What Doesn't Work

The honest account must include the failures, because the failure rate is the industry's dirty secret. Most digitalisation initiatives in legacy industry do not deliver the promised returns. McKinsey and other consultancies have repeatedly found that a large share of industrial IoT and analytics projects stall at the pilot stage or fail to scale — the famous "pilot purgatory." The reasons are consistent and identifiable.

The first is the data problem. Industrial data is messy: different vintages of machines, incompatible protocols, missing historians, bad tags, and signal noise. Cleaning it consumes most of the effort and most of the budget, and the organisations that underestimated this never recover. The second is the skills problem: analytics requires people who understand both the data and the process, and those people are rarer than either data scientists or process engineers. The third is the organisational problem: digital insight threatens existing power structures — the maintenance manager whose judgment is questioned, the operator whose expertise is devalued, the department whose budget is reallocated — and organisations resist what they cannot control. The fourth is the expectation problem: companies expect the platform to produce insight, when insight requires the same hard, domain-specific design work that produced the physical product in the first place.

None of these failures is a technology failure. They are design failures — failures to design the digital initiative around the people, the processes, and the specific problem. The companies that succeed treat digital-physical integration the way they treat the physical product itself: as something to be designed, iterated, tested, and refined against real use, not installed and switched on.

The Frontier as Competitive Terrain

The digital-physical frontier is not a technology trend. It is the terrain on which the next industrial age will be fought. The companies that can see their assets in real time, simulate their future, and feed the evidence back into design will out-compete the companies that cannot, in every dimension that matters: reliability, efficiency, speed of iteration, cost, and the ability to retain the knowledge of a retiring workforce. The tools are no longer exotic. The challenge is design — deciding what to digitise, what to connect, and what to do with the result.

Toolkit Summary

- The digital twin is a two-way mirror, not a static model: live sensor data, an engineering model, and simulation fused into continuous insight about a physical asset.

- The canonical success is Rolls-Royce engine health monitoring: predicted, not discovered, failures; maintenance scheduled around operations; every engine's model refined by every flight.
- MindSphere and Predix show the difference between anchored and abstract digitalisation: platforms succeed when tied to specific assets and problems, and fail when sold as general-purpose bets awaiting users.
- Retrofitting IoT to legacy machinery — vibration, oil, current, thermal sensing — is cheap and transformational, but only when insight is wired into work orders rather than admired in dashboards.
- The digital thread closes the design feedback loop: traceability, compliance, and materials passports become by-products of normal operation, and design decisions are made from evidence instead of assumptions.
- Most industrial digitalisation fails — pilot purgatory — for data, skills, organisational, and expectation reasons that are design failures, not technology failures.
- Digital-physical integration is a design discipline: decide what to digitise, who acts on it, and how insight returns to the design of the next product.

Chapter 4: Sustainability as a Design Imperative

FOR MOST OF INDUSTRIAL HISTORY, SUSTAINABILITY WAS AN EXTERNALITY — A COST IMPOSED ON SOCIETY AND NATURE BY THE ACT OF PRODUCTION, BORNE BY NEITHER THE PRODUCER NOR THE BUYER, AND ADDRESSED, IF AT ALL, AT THE END OF THE PIPE. POLLUTION WAS TREATED AT THE SMOKESTACK. WASTE WAS TREATED AT THE LANDFILL. EMISSIONS WERE TREATED, GRUDGINGLY, AT THE POINT OF REGULATION. THE PRODUCT ITSELF — ITS MATERIALS, ITS ENERGY INTENSITY, ITS REPAIRABILITY, ITS FATE AFTER SALE — WAS ALMOST NEVER THE SUBJECT OF DESIGN ATTENTION. THAT ERA IS OVER. SUSTAINABILITY HAS MOVED FROM THE PERIPHERY OF INDUSTRIAL PRACTICE TO ITS CENTRE, AND IT HAS DONE SO NOT AS A MATTER OF ETHICS BUT AS A MATTER OF PHYSICS, REGULATION, AND MONEY. IT IS NOW A DESIGN CONSTRAINT AS FUNDAMENTAL AS STRENGTH, COST, OR SAFETY — AND IT IS RESHAPING WHAT LEGACY INDUSTRY BUILDS AND HOW IT BUILDS IT.

The reasons are not mysterious. The industrial sectors that underpin the Western economy — steel, cement, chemicals, aluminium, refining, heavy transport — are also the hardest to decarbonise and the most emissions-intensive. They are the "hard-to-abate" sectors, responsible for a disproportionate share of industrial

emissions precisely because their processes require high temperatures, chemical reactions, and energy densities that electricity and efficiency alone cannot easily replace. The pressure on them comes from three directions at once. Regulators are converting climate targets into binding product standards. Customers — especially the large purchasers who dominate industrial supply chains — are demanding verified low-carbon inputs. And capital markets are pricing transition risk into the cost of borrowing. Any of these pressures alone would force change. Together, they make sustainability a condition of doing business, and therefore a design problem: the question is no longer whether to decarbonise, but how to redesign products, processes, and business models so that they work within planetary and regulatory limits.

Circularity as a Design Discipline

The deepest shift is from a linear to a circular conception of the product's life. The linear model — take materials, make product, use product, discard product — treated the end of life as the end of the designer's responsibility. The circular model treats the end of life as a design input: the product must be capable of being repaired, refurbished, remanufactured, or disassembled into materials that re-enter production. This is not recycling-as-afterthought, which in heavy industry usually means downcycling — steel scrap into lower-grade steel, concrete into aggregate, composite waste into filler. It is

design-for-circularity: making decisions at the drawing board that determine whether a product's materials can be recovered at high value decades later.

Designing for disassembly is the discipline with the most immediate payoff. It means joints that can be undone rather than welded, fasteners standardised rather than proprietary, components arranged for access rather than for assembly-line speed alone, and hazardous materials isolated in modules that can be removed before the rest is recycled. The automotive industry has been pushed in this direction by regulation for years — the EU's End-of-Life Vehicles directive requires that a large share of each car's mass be recoverable, which forced designers to think about recyclability as a specification. The pattern is now spreading to every heavy sector. Wind turbine blades, once designed to be glued and cured into a single inseparable unit, are being redesigned for separability, with manufacturers like Siemens Gamesa developing recyclable blade resins precisely because the industry can see that hundreds of thousands of tonnes of blade material will reach end of life in the coming decade. Industrial equipment is following: manufacturers of machinery and electronics are redesigning products so that components can be removed, tested, and reused, rather than discarded with the assembly.

Design for repair is the companion discipline, and it is where the design constraint bites hardest against established business models. A product designed to be repaired is a product designed not to be replaced on a

schedule. For industries that historically derived a comfortable revenue stream from spare parts and replacement cycles, repairability looks like a threat. Regulation is removing the choice. The EU's right-to-repair legislation, adopted as part of the broader ecodesign framework, obliges manufacturers to make spare parts available for defined periods, to publish repair information, and to design products so that repairs are physically possible without destroying the product. The ecodesign regulation itself — the EU's umbrella framework for product sustainability requirements — is systematically extending from consumer goods into industrial equipment: energy-related products, batteries, and soon a widening range of industrial categories must meet minimum durability, repairability, and recyclability criteria to be sold in the European market. The significance for legacy industry is hard to overstate: the world's largest single market is now writing circularity into the design specification of industrial products.

The Hard-to-Abate Core

The most intractable design problems are the materials that civilisation is literally built from: steel and concrete. Together with chemicals and aluminium, they account for the bulk of industrial emissions, and they cannot be decarbonised by switching fuel alone, because their emissions come from chemistry as much as combustion. Making steel in a blast furnace emits carbon dioxide from

the reduction of iron ore by coke — a chemical necessity, not an energy choice. Making cement emits carbon dioxide from the calcination of limestone — again, chemistry. These sectors are the reason industrial decarbonisation is hard, and they are where the design thinking of the next industrial age will be tested most severely.

The strategies are emerging in parallel. The first is process change: hydrogen-based direct reduction of iron, electric arc furnaces fed by scrap and increasingly by direct-reduced iron, and carbon capture on the unavoidable process emissions. The second is material efficiency: designing structures and products that use less steel and less concrete in the first place — optimised sections, higher-strength grades, design codes that reward rather than penalise material savings. The third is the substitution of the material itself: timber and engineered wood products moving into mid-rise construction, low-carbon cement chemistries, steel produced from scrap in electric furnaces at a fraction of the primary-route footprint. The fourth is the transformation of the asset base: because steel plants and cement kilns last for decades, the plants being designed today determine the emissions of 2050. Retrofitting existing plants with new capture technology, or replacing them with fundamentally different processes, is a design decision of the highest order — one that is currently being made, plant by plant, in every industrial region of the West.

None of this is cheap, and the honest framing is that the transition will be paid for by higher prices for the most emissions-intensive materials — which is precisely the point. When steel and concrete become more expensive, the design incentives throughout the economy shift: lighter structures, longer-lasting products, reused components, and materials chosen for their whole-life carbon rather than their first cost. The same logic applies inside the sectors themselves. The aluminium industry, which has a viable low-carbon route in hydropower-fed smelting and recycled metal, is being redesigned around those advantages; the chemical industry, which has the most complex and least tractable emissions profile of all, is being pulled toward recycled feedstocks, electrified processes, and products designed to be recovered rather than dispersed. Carbon pricing, where it is credible, amplifies all of these incentives. The design community's role is to make the response to those incentives elegant rather than grudging — to design the low-carbon version as the better product, not the worse one with a green label.

Design Strategies for Decarbonisation

The design strategies available to legacy industry divide into four families, and the most sophisticated companies are pursuing all of them.

The first is energy and process redesign: electrifying heat, recovering waste heat, optimising process chemistry, and — where the process fundamentally requires it — capturing carbon at the source. This is the territory of the process engineer, but it is design in the truest sense: the redesign of the industrial process itself as an integrated system rather than a collection of unit operations.

The second is product redesign for the low-carbon world: the gas turbine designed to burn hydrogen, the engine designed for sustainable fuels, the vehicle designed for electrification, the chemical plant designed around recycled feedstocks. The companies winning the energy transition are the ones that treat it as a product portfolio problem. A gas turbine maker that designs its next generation for 100 percent hydrogen combustion is not making a greener version of the old product; it is designing the product the market will demand in twenty years.

The third is design for the second-life economy. Industrial equipment — turbines, pumps, transformers, switchgear, entire production lines — is long-lived, and its second life is now a design requirement. Equipment designed for remanufacture can be rebuilt to like-new condition at a fraction of the carbon cost of new production, and the market for this is growing rapidly as customers seek to decarbonise their procurement. The companies that design for remanufacture — with modular architectures, standardised interfaces, and condition data

built in — will capture a growing share of a market that is being created by regulation and by customer demand. The companies that don't will watch their installed base become someone else's raw material.

The fourth is transparency: designing the product's environmental data into its digital record. The materials passport — the digital document that records what a product is made of, its carbon footprint, its repairability, and its disassembly instructions — is moving from pilot to requirement. The EU's battery regulation has made battery passports mandatory, and the logic is spreading to construction products and industrial equipment. A product with a passport is a product that can be traded, refurbished, and recycled in a functioning secondary market. A product without one is a product that ends its life as waste, because nobody can establish what it is worth recovering.

The Constraint That Liberates

There is a temptation to describe all of this as burden — regulation, cost, constraint. The more accurate framing, and the more useful one for design, is the opposite. Sustainability constraints are doing for industrial design what safety regulation did for aviation and emissions regulation did for the car: forcing a step-change in sophistication that the industry would not have taken on its own. The car became a better product —

more efficient, more reliable, more sophisticated — because regulators forced engineers to think harder. The same is happening now, at industrial scale.

The evidence is already visible. The wind industry's recyclable blades. The steel industry's electric-arc revolution and its hydrogen pilots. The chemical industry's experiments with electrified cracking and bio-based feedstocks. The construction industry's engineered timber and low-carbon concrete. In each case, the constraint did not shrink the industry's ambition; it redirected it. The companies that treat sustainability as a design imperative — as the specification for the next generation of products, processes, and business models — are the ones designing the industrial future. The ones that treat it as compliance are the ones being designed for.

Toolkit Summary

- Sustainability is now a hard design constraint — driven by regulation, customer procurement demands, and capital markets — not an ethical extra or an end-of-pipe problem.
- Circular design means designing for disassembly, repair, and high-value recovery at the drawing board; the EU's right-to-repair and ecodesign regulations are writing these requirements into the legal specification of industrial products.

- The hard-to-abate core — steel, cement, chemicals, aluminium — requires process change, material efficiency, substitution, and asset-base decisions that lock in 2050 emissions today.
- Four design strategy families: energy/process redesign, product redesign for the low-carbon market, design for the second-life economy (remanufacture), and environmental transparency via materials passports.
- Materials passports turn end-of-life products into tradeable assets; products without them end their lives as waste.
- Sustainability constraints are forcing a step-change in industrial sophistication — the same dynamic by which safety and emissions regulation transformed aviation and the car.
- The winners treat sustainability as the specification for the next product generation; the losers treat it as compliance.

Chapter 5: Materials and the Next Revolution

EVERY INDUSTRIAL ERA IS, UNDERNEATH EVERYTHING ELSE, A MATERIALS ERA. THE IRON AGE, THE STEEL AGE, THE AGE OF ALUMINIUM, THE AGE OF SILICON — EACH TRANSFORMATION IN WHAT INDUSTRY CAN BUILD HAS BEGUN WITH A TRANSFORMATION IN WHAT INDUSTRY CAN MAKE THINGS FROM. THE NEXT INDUSTRIAL AGE IS NO EXCEPTION. THE MATERIALS LANDSCAPE IS SHIFTING BENEATH THE FEET OF LEGACY INDUSTRY, AND THE COMPANIES THAT UNDERSTAND HOW MATERIALS CHOICE DRIVES DESIGN WILL BUILD THE FUTURE; THE ONES THAT TREAT MATERIALS AS A PROCUREMENT CATEGORY WILL BE STRANDED BY IT.

Three forces are reshaping the landscape simultaneously. The first is performance: new materials — advanced composites, superalloys, engineered ceramics, high-entropy alloys — offer properties that old materials cannot match, and they are moving out of the laboratory and into production at scale. The second is sustainability: the carbon, toxicity, and end-of-life profile of a material is now a design input, and recycled and bio-based materials are shifting from niche to mainstream. The third is digitalisation: materials are becoming data — specified, traced, and traded on the strength of their records. A material without a passport is becoming a material without a future.

How Materials Drive Design

The relationship between materials and design is not a one-way street. It is a negotiation. A material's properties — strength, stiffness, density, conductivity, corrosion resistance, fatigue life, cost, availability, recyclability — define the space of what can be designed, and design choices in turn define what materials are demanded at scale. The classic illustration is the relationship between aluminium and the aircraft. The aluminium alloys of the 1930s made the modern airliner possible; the economics of the airliner then drove the development of aluminium alloys for a century. When composites arrived, they did not just offer a lighter alternative — they changed what an aircraft could be. The Boeing 787 and Airbus A350 are not aluminium aircraft with composite skins. They are aircraft designed around composite structures, with different fuselage pressurisation limits, different maintenance regimes, and different manufacturing economics, because the material's properties allowed — and demanded — a different design.

This is the pattern that matters for legacy industry. When a material changes, the design does not change by substitution; it changes by reimagining. The designer who thinks "replace the steel part with a composite part" misses the point. The designer who asks "what can this part become, and what can the product become, if the material constraints are different?" is doing the work of the next industrial age.

Additive Manufacturing: From Prototype to Production

No material technology has been more overhyped and more misunderstood than additive manufacturing. For two decades, 3D printing was presented as a technology that would replace conventional manufacturing — a promise that has not come true and will not, because it misunderstands the economics. Machining, casting, forging, and forming are extraordinarily good at making parts in quantity, and they are getting better. Additive manufacturing's real role is different and, in its way, more important: it is the technology for parts that cannot be made any other way, and for the production economics that quantity manufacturing cannot serve.

The industrial applications that actually work share a consistent pattern. They are high-value, low-volume, geometrically impossible by other means, or needed in quantities too small to justify tooling. In aerospace, additive manufacturing has moved decisively from prototyping to production: GE Aviation's fuel nozzles for the LEAP engine — a single part replacing an assembly of twenty, with internal cooling channels that could not be cast or machined — are produced by the hundreds of thousands. Safran, Rolls-Royce, and MTU produce flight-critical titanium and nickel-alloy parts by laser powder-bed fusion and electron-beam melting. In medical devices, the pattern is identical: patient-specific implants and surgical guides that are, by definition, one-offs. In heavy

machinery, the pattern is emerging: on-demand spare parts, printed at the point of need, eliminating warehouses of obsolete stock. In tooling, conformal cooling channels inside injection moulds and die-casting dies cut cycle times dramatically — a small design change with large production economics.

The design lesson of additive manufacturing is the discipline it imposes. The technology rewards design freedom — topology-optimised structures, lattice interiors, parts consolidated from assemblies — and punishes the transfer of conventional designs unchanged. The companies that capture value from it are the ones that redesign parts for the process, and the design tools are now mature: generative design software explores thousands of material-efficient geometries that no human designer would conceive, and the printer turns the chosen geometry into metal. The companies that merely print their old designs are paying a premium for worse parts.

Carbon Fibre: The Journey to Mass Production

Carbon fibre's trajectory is the story of advanced materials in miniature. Developed for aerospace and defence — where its combination of stiffness, strength, and lightness justified almost any cost — it spent decades as an exotic material priced by the kilogram like a precious metal. The journey from aerospace to mass

production was driven by the search for volume applications that could pull costs down the learning curve, and the vehicle that did it was the mass-market car. BMW's i3 and i7 programmes, with their carbon-fibre passenger cells, were the industrial commitment that built the capacity and process knowledge; Tesla's use of carbon composites in the Model S and Roadster pushed the material further into the mainstream.

The result is a material in transition. Carbon fibre is still expensive relative to steel and aluminium, but its cost has fallen steadily, its manufacturing processes — automated fibre placement, out-of-autoclave curing, recycled fibre — have matured, and its applications have widened from aerospace and racing cars to wind turbine blades (the single largest volume market), pressure vessels for hydrogen storage, and construction reinforcement. The design lesson is about the interplay of cost and volume: materials do not become cheap and then find applications; applications are designed that pull materials down the cost curve. The wind industry did for carbon fibre what BMW did for automotive composites — and the hydrogen economy is now doing it again, because high-pressure hydrogen storage is essentially a carbon-fibre design problem.

Recycled and Bio-Based Materials

The second force reshaping the landscape is the demand for materials with a lower environmental footprint, and it is producing two distinct streams of innovation. The first is high-quality recycling: materials recovered at sufficient purity and consistency to re-enter demanding applications. The steel industry's electric arc furnace route — scrap-fed, radically lower carbon than the blast furnace — is the mature exemplar; aluminium recycling, at a fraction of the energy cost of primary production, is another. The frontier is in materials that were never designed to be recycled: composites, which are being addressed through pyrolysis and solvolysis to recover fibre; plastics, through chemical recycling that breaks polymers back to monomers; and batteries, through a rapidly scaling recycling industry that recovers lithium, nickel, cobalt, and graphite. The design implication is direct: materials that can be recycled to specification are materials with a future; materials that cannot are becoming design liabilities.

The second stream is bio-based materials: polymers from biomass, fibres from agricultural waste, chemicals from fermentation, and construction materials from engineered biological sources. These are not a return to pre-industrial materials; they are industrial materials with a different feedstock, and they are being designed into products where their properties — and their carbon story — earn their place. The design lesson is that bio-based does not mean worse; it means different, and different

requires design. A composite made from flax or hemp fibres, a polymer from corn or captured carbon, a construction block from mycelium — each requires the product to be designed around its properties rather than substituted into an existing design.

Battery Materials and the Electrification of Industry

The most strategically consequential materials story of the next industrial age is the one happening inside batteries, because electrification is the master process of industrial decarbonisation, and batteries are its material bottleneck. The scale of the demand is hard to overstate: the battery supply chain — lithium, nickel, cobalt, graphite, and the cathode and anode materials made from them — is being asked to grow from a niche industry to the scale of the oil industry within a few decades. Every industrial region in the West is now racing to build refining, precursor, cathode, and cell capacity, having discovered that the entire clean-economy build-out — vehicles, grid storage, industrial equipment, defence systems — depends on it.

The design consequences ripple outward. Battery chemistry choices drive vehicle and equipment design: cell format, pack architecture, thermal management, and charging capability are all negotiated with the materials supply chain. The design of the pack determines the design of the second life — a battery retired from a

vehicle retains value for grid storage, and a pack designed for second life is a different product from one designed for disposal. And the design of the battery determines its recyclability: the "design for recycling" movement in batteries — cells that can be separated, chemistries that can be recovered, packs that can be disassembled — is now a competitive front, because the industry can see that the materials in deployed batteries will be the feedstock of the future, and the companies that can recover them will control the resource. The EU's battery regulation, with its mandatory passports and recycled-content requirements, is turning these design choices into legal requirements.

Materials Passports and Traceability

Which brings the third force to the foreground: materials are becoming data. The materials passport — the digital record of a material's composition, provenance, carbon footprint, and recovery potential — is the connective tissue of the circular economy, and it is moving from concept to requirement. Without a passport, a product's materials are unknowable at end of life: what they are, whether they are hazardous, what they are worth recovering. With a passport, the material becomes a tradeable asset with a documented identity. The construction industry — where buildings are the largest material stock on earth — is pioneering this, with digital registries of building materials being piloted across

Europe, driven by the EU's push for digital building logbooks. Industry is following: equipment passports, battery passports, and component-level traceability are becoming standard practice in the sectors that expect to recover their materials.

The deeper significance of traceability is that it makes the materials economy legible. A supply chain that can document where every material came from, what it contains, and what it will become is a supply chain that can be optimised, regulated, and trusted. The companies that build traceability into their products — by design, from the start — will be the ones that can guarantee their materials, prove their carbon claims, and capture the value of recovery. The companies that don't will find their materials increasingly difficult to sell into markets that demand proof.

The Material Future Is Designed

The materials revolution is not a list of new substances waiting to be adopted. It is a design agenda. Every material decision is a design decision: what the product can be, how it is made, what it costs over its whole life, and what happens to it at the end. The next industrial age will be built from materials that did not exist in the last one, and from old materials used in new ways — and the difference between winning and losing will be the quality of the design thinking applied at the interface between material and product.

Toolkit Summary

- Every industrial era is a materials era; materials choice is the deepest design decision a company makes, and changing materials means reimagining the product, not substituting parts.
- Additive manufacturing's real role is production of parts that cannot be made otherwise — GE Aviation's LEAP fuel nozzles, patient-specific implants, on-demand spares — not replacement of conventional manufacturing at scale.
- Carbon fibre's trajectory shows the cost-volume flywheel: applications (BMW, wind blades, hydrogen storage) pull materials down the cost curve; materials don't get cheap before they get adopted.
- Recycled and bio-based materials are mainstream design inputs, and recyclability to specification is becoming a design requirement, not a marketing claim.
- Batteries are the material bottleneck of electrification; pack architecture, second-life design, and design-for-recycling are competitive fronts, now codified in regulation like the EU battery passport.
- Materials passports and traceability turn end-of-life materials into tradeable assets and make supply chains legible; materials without records are becoming design liabilities.

- The winners design from material capability forward; the losers treat materials as a procurement category.

Chapter 6: Automation, AI, and Human-Centered Design

NO SUBJECT IN INDUSTRIAL STRATEGY IS MORE LOADED THAN AUTOMATION. IT PROMISES THE PRODUCTIVITY THAT LEGACY INDUSTRY DESPERATELY NEEDS — AND THREATENS THE EMPLOYMENT THAT LEGACY INDUSTRY DISPROPORTIONATELY PROVIDES. IT IS SOLD AS THE ANSWER TO THE SKILLS SHORTAGE — AND ACCUSED OF DESTROYING THE SKILLS THAT REMAIN. IT IS PRESENTED AS INEVITABLE, AND RESISTED AS INHUMAN. BOTH THE PROMISE AND THE THREAT ARE REAL, WHICH IS WHY THE DESIGN QUESTION IS THE DECISIVE ONE: AUTOMATION THAT IS DESIGNED AROUND PEOPLE, PROCESSES, AND SPECIFIC PROBLEMS BECOMES A COMPETITIVE WEAPON; AUTOMATION THAT IS INSTALLED AS A SUBSTITUTE FOR THINKING BECOMES AN EXPENSIVE WAY TO FAIL.

The first thing to clear away is the fantasy version of the debate. The fully autonomous factory — lights out, human-free, self-optimising — exists, but it is a narrow special case, not a template. It works for high-volume, low-variety, predictable production of simple things: certain consumer electronics, certain packaging, certain assembly operations. It does not work — and will not work for decades — for the kind of work that dominates legacy industry: low-volume, high-variety, expert-

dependent work on complex assets, much of it in environments that change faster than any fixed automation can be reprogrammed. A shipyard, a chemical plant turnaround, a defence fabrication shop, a maintenance hangar, a nuclear outage — these are not lights-out environments. They are environments where the human is the most sophisticated sensing, reasoning, and decision-making system present, and where automation's job is to make that human more capable, not to replace them.

The Automation Reality in Legacy Industry

The reality on the ground is more interesting than either the utopian or dystopian story. In the sectors this book is about, automation is being adopted selectively, unevenly, and almost always in the presence of people. The welding robot in the fabrication shop works alongside welders on other stations. The automated guided vehicles in the warehouse coexist with forklift drivers. The exoskeleton on the assembly line augments the worker who wears it. The predictive-maintenance algorithm supports the maintenance planner who schedules the work. This is not a transitional state on the way to full autonomy; it is the endpoint. The most productive form of automation for most industrial work is augmentation, not replacement.

The reasons are economic as much as human. Full automation is only rational when the volume justifies the capital and the process is stable enough to automate. Legacy industry's work is frequently neither. A defence prime building platforms in tens rather than thousands, a heavy-machinery maker with hundreds of variants, a chemical plant with processes that must be adapted to feedstocks and conditions — these operations cannot amortise fixed automation, and they benefit instead from flexible automation that amplifies the human workforce they already have. The strategic question for every legacy company is not "how do we automate this job?" It is "what is the right division of labour between people and machines for this work, at this volume, in this environment?" That is a design question.

Cobots and the Middle Ground

The most visible symbol of the middle ground is the collaborative robot — the cobot. Where the industrial robot of the twentieth century was a caged hazard, separated from people by fences and light curtains, the cobot is designed to work in the same space as people, sensing their presence, yielding when touched, and taking on the tasks that people do worst: the repetitive, the heavy, the ergonomically destructive, the precise-to-a-fraction-of-a-millimetre-and-boring.

The design logic of cobots is worth spelling out, because it inverts the automation logic of the past. A traditional robot replaced a worker; a cobot relieves one. The cobot does the parts of the job that damage human bodies and brains — the thousandth identical lift, the task requiring sustained force, the operation that demands concentration past the point of fatigue — and the human does the parts that defeat robots: judgment, adaptation, quality assessment, and the handling of exceptions. In welding, the cobot holds the torch steady through the long seam while the welder sets up, inspects, and handles the awkward corners. In machining, the cobot loads and unloads the machine tool while the machinist manages the process. In assembly, the cobot presents parts and applies force while the assembler does the fitting and the fastening. In every case, the unit of production is not the robot or the worker alone; it is the pair.

This is why the companies that treat cobots as cheap robots — install them, program them, and let them replace workers one-for-one — miss the point and often fail. The cobot's value is in the redesign of the work itself: the work cell rearranged so that human and machine each do what they do best, with the interface between them designed as carefully as the machinery. That redesign is a design task in the fullest sense, and the companies that do it well report the outcomes that the automation industry promises: higher throughput, better

quality, lower injury rates — and, critically, higher retention, because the work stops destroying people's bodies.

Where AI Genuinely Helps Industrial Design

The role of artificial intelligence in this picture is frequently misunderstood, and the misunderstanding is costly. The useful framing is not "AI replaces the designer" or even "AI assists the designer." It is that AI changes the economics of exploration: it lets designers search vastly larger spaces of possible solutions than human effort alone could cover, and it lets them find solutions that human intuition would never generate.

Generative design is the clearest example. Given the design requirements — the loads, the constraints, the manufacturing process, the materials — generative design software produces thousands of candidate geometries, many of them organic, counterintuitive, and better than anything a human would have drawn. The designer's role becomes curation: setting the requirements, evaluating the candidates, and choosing the solution that fits the broader system. This is not automation of design; it is augmentation of design, and it is already in production use — in brackets and fittings for aircraft and vehicles, in optimised structural components

for machinery, in heat exchangers and impellers where fluid dynamics rewards shapes no drafting hand could produce.

The second genuine strength of AI is optimisation of existing designs and processes. Machine learning models trained on operational data can find the operating point of a plant that minimises energy per tonne, the maintenance interval for a fleet that minimises total cost, the scheduling of a job shop that maximises throughput. The third is simulation acceleration: AI models trained on high-fidelity physics simulations can approximate those simulations in milliseconds, enabling design iteration at a speed that turns "can't test that" into "tested a thousand variants." In materials, this is transforming alloy and chemistry development: instead of a decade of experimental trial and error, candidate materials are screened *in silico* and the few survivors are made and tested.

The discipline that makes all of this work is the same discipline that makes every other digital tool work in industry: the AI must be anchored to a specific problem, trained on real data from the actual operation, and integrated into a workflow where a human is accountable for the result. AI systems that are fed to organisations as general-purpose platforms — the "buy an AI platform and wait for value" pattern — fail at the same rate and for the same reasons as the digital platform bets of the past decade. AI is not a strategy; it is a design material.

The Human-Machine Interface Challenge

The interface between people and automated systems is where the best automation goes to die, and it is the most neglected design problem in industry. The pattern is universal: a plant installs a sophisticated automation or analytics system, and the people who operate it — the operators, technicians, and planners whose judgment the system is supposed to support — either don't trust it, don't understand it, or are actively undermined by it. The result is expensive hardware running in manual override, or brilliant dashboards that nobody acts on.

The design failures are identifiable and preventable. The first is opacity: systems that present conclusions without the reasoning behind them, leaving operators unable to decide when to trust the machine and when to override it. The fix is explainability by design — the system presents its reasoning, its confidence, and the evidence it is acting on, and it is designed to be argued with. The second is mistrust born of error: every automation system makes mistakes, and if the system cannot acknowledge uncertainty, one mistake destroys a year of trust. The fix is designing for graceful degradation — the system knows what it does not know, and escalates to the human rather than confidently continuing. The third is deskilling: systems that take the human out of the loop entirely, so that when the system fails — and every system fails — there is no human left who understands the process well enough to take over. The famous automation paradox: the more reliable the automation,

the less practiced the human, and the worse the human performs at the rare moments when they are needed most. The fix is designing automation that keeps the human in the loop, practicing the skills, so that the human is a capable backup rather than a ceremonial observer.

Designing Automation That Works With, Not Instead Of, Workers

The design principle that ties all of this together is deceptively simple: automation should be designed to make the worker's job better, not to make the worker unnecessary. The phrasing matters, because the two goals produce different designs. Automation designed to replace people is optimised for the absence of people: it removes the human interface, eliminates the need for human judgment, and treats the worker as an obstacle to be designed around. Automation designed to work with people is optimised for the presence of people: it is legible, explainable, error-tolerant, and it gives the human the information and the authority to make the final call.

The difference shows up in measurable outcomes. Automation designed around workers gets adopted — and adopted automation is what delivers the productivity gains. Automation designed against workers gets resisted, bypassed, sabotaged, and eventually written off. The resistance is not Luddism; it is rational response to a

design that fails to account for the people in the system. The plants and firms that succeed treat the workers as the system's most valuable component, and design automation to amplify them.

It would be dishonest to present augmentation as the whole story. There is a dark side, and the design community that ignores it forfeits its credibility. Even in the augmentation model, automation eliminates jobs — the loading and unloading tasks, the repetitive welds, the data entry, the inspection roles that algorithms now perform. The jobs that remain are more skilled and better paid, but the workers who held the eliminated jobs are not automatically the workers who get the better ones. And automation that removes the need for practiced judgment — the operator who no longer balances the plant by feel — erodes exactly the expertise that industry needs for its rare emergencies. The distributional question is unavoidable: the gains from automation accrue to capital, the costs to labour, unless the design of the system — and the policy around it — deliberately shares both.

None of this is an argument against automation. It is an argument for designing it honestly, with the costs acknowledged and the trade-offs made explicit rather than hidden. The industry that automates while pretending there are no losers will reap the resistance; the industry that automates while designing for the

people it affects — retraining, redeploying, and redesigning work around them — will get the adoption and the productivity.

Human-Centered Automation Frameworks

The good news is that the frameworks for doing this properly exist and are proven. Human-centred automation — the design practice that puts the operator's cognition, workload, and authority at the centre of the system design — has a long pedigree in aviation, where it produced the modern cockpit philosophy: the machine flies, the pilot commands, and the design of every system respects the division. That philosophy transferred to process industries through the work of human-factors engineers on control-room design, and it is now the standard to which the best industrial automation is held.

The practical framework has four elements. First, function allocation: decide, deliberately and per task, what the machine does and what the human does, based on what each does best — not on what the technology can technically do. Second, authority and accountability: design the system so the human has the authority to override, and make the human accountable for the outcome, which forces the system to earn and maintain the human's trust. Third, transparency: the system must be able to show its work — its state, its reasoning, its confidence — in a form the operator can read under time pressure. Fourth, skill preservation: design the system to

keep the human practiced in the skills that would be needed if the system failed, through training, simulation, and periodic handover of control.

Legacy industry has an advantage it rarely recognises: it still has the experts. The retiring generation, the experienced operators and maintainers, the people whose judgment the automation is meant to support — they are the raw material of human-centred automation design. The companies that involve them in designing the automation — that ask them what the machine should do, what it should never do, and how it should explain itself — will build automation that works. The companies that design automation in an engineering office and impose it on the shop floor will build monuments to wasted investment. The next industrial age will not be automated. It will be augmented — by machines designed, in every sense, around the people who make industry work.

Toolkit Summary

- The lights-out factory is a narrow special case; the dominant model for legacy industry is augmentation — automation that makes expert workers more capable, not automation that replaces them.

- Cobots work when the work cell is redesigned around the human-machine pair (each doing what they do best), not when they are installed as cheap one-for-one replacements.
- AI's genuine value in design is exploration at scale — generative design, optimisation on real operational data, and simulation acceleration — with the human curating requirements and choices; AI is a design material, not a strategy.
- The human-machine interface is where automation fails: opacity, one-mistake-trust-erosion, and deskilling are design failures with known fixes — explainability, graceful degradation, and keeping the human in the loop.
- Design automation that makes the worker's job better, not the worker unnecessary; adopted automation is the only automation that delivers productivity.
- The dark side is real — displacement, skill atrophy, and uneven distribution of gains — and must be designed for honestly, with trade-offs explicit.
- Human-centred automation frameworks — function allocation, authority and accountability, transparency, skill preservation — are proven in aviation and process control; legacy industry's retiring experts are the raw material for designing automation that works with people.

Chapter 7: Designing for the Worker of the Future

THE MACHINIST IS SIXTY-TWO YEARS OLD. HE HAS OPERATED THE SAME CNC LATHE FOR THIRTY-ONE YEARS, AND HE CAN TELL YOU, FROM THE SOUND OF THE SPINDLE, WHEN THE BEARINGS ARE GOING TO FAIL A WEEK BEFORE THE VIBRATION SENSORS PICK IT UP. HE CAN SET UP A JOB IN HALF THE TIME THE ENGINEERS' PROCESS SHEET ALLOWS. HE HAS NEVER WRITTEN A LINE OF CODE, AND HE IS RETIRING IN FOURTEEN MONTHS. WHEN HE LEAVES, THE COMPANY WILL NOT JUST LOSE AN OPERATOR. IT WILL LOSE A DIAGNOSTIC SYSTEM, A TRAINING MANUAL, AND A QUALITY CONTROL DEPARTMENT, ALL WRAPPED IN A STEEL-TOED BOOT.

This is the quiet crisis at the heart of Western legacy industry. Every forecast about automation, AI, and the factory of the future presumes a workforce that will operate it. But that workforce is retiring faster than it is being replaced, and the expertise leaving through the factory gate is not recorded anywhere. It lives in hands, ears, and intuitions developed over decades. The design question of the next industrial age is therefore not only about machines. It is about the humans who run them — the ones leaving, the ones staying, and the ones we have not yet persuaded to come.

The Knowledge Transfer Problem

The demographics are unforgiving. In the United States, the median age of a manufacturing worker has climbed past forty-four, and more than a quarter of the workforce is over fifty-five. In Germany, the industrial heartland of Europe, the situation is starker still: the country's engineering sector faces a shortage of skilled workers measured in the hundreds of thousands, with the gap widening every year as the baby-boom generation exits. The European Commission's own analysis projects that the bloc will lose millions of workers to retirement over the coming decade, with manufacturing among the hardest-hit sectors. The phrase "silver tsunami" has become a cliché in HR presentations, but the reality behind it is not. The largest cohort of industrial expertise in Western history is walking out the door, and most companies have no systematic plan for what it knows.

The problem is not that this knowledge is secret. It is that it is tacit. The machinist's ability to hear bearing wear is not written in any manual. The welder's feel for the correct puddle of molten metal is not in any weld procedure specification. The plant manager's judgment about which supplier will actually deliver on time is not in any ERP system. Tacit knowledge, by definition, resists documentation. It is acquired through years of embodied practice, and it is lost the moment it is not deliberately transferred.

Design has a role here that goes far beyond writing better procedures. The tools of the digital industrial age — sensors, digital twins, machine learning — can capture the outputs of tacit knowledge and make them explicit. A vibration signature is the machinist's ear, made visible. An expert-system playbook is the plant manager's judgment, made shareable. But capturing knowledge is only half the problem. The other half is creating the conditions under which transfer actually happens: the older worker mentoring the apprentice, the veteran operator co-designing the new control interface rather than having it imposed on them, the retiring engineer whose last project is not a handover document but a living simulation that trains her replacement.

The companies that treat this as a design challenge — designing the transfer rather than merely the documentation — will survive the demographic cliff. Those that treat it as an HR checklist will find themselves, a decade from now, with gleaming automated factories and no one who understands how to keep them running.

Designing for Bodies That Age

There is a second, more immediate design problem: the bodies doing industrial work are older than they have ever been, and industrial workplaces were largely designed for twenty-five-year-olds. This is not a matter of sentiment; it is a matter of safety and productivity. Fatigue, reduced grip strength, diminished eyesight, and

slower recovery from injury are not character flaws. They are engineering parameters. Good design treats them as such.

Ergonomics has been a formal discipline for decades — the ISO 9241 family of standards on human-system interaction, the biomechanics research that underpins modern workstation design, the lifting guidelines that have reshaped material handling. But in many legacy plants, the equipment predates all of it. A control panel mounted at a height comfortable for a six-foot operator in 1985 is a strain for a five-foot-five operator in her sixties today. A hand tool that requires a crushing grip accelerates arthritis. A display with twelve-point type is illegible to eyes that need fourteen. These are not exotic problems. They are the daily texture of aging industrial work, and they are all design decisions, made badly by default.

The emerging assistive technologies are genuinely promising, and the design community should treat them not as novelties but as accessibility infrastructure. Passive and active exoskeletons — wearable frames that offload the strain of overhead work and heavy lifting — have moved from research labs into commercial use in automotive and aerospace assembly, where they demonstrably reduce fatigue and injury rates. Vision-assist systems, from simple magnification to AR headsets that overlay part numbers and torque specifications, compensate for aging eyes while simultaneously reducing error rates. Powered material-handling aids, collaborative

robots that take the heavy end of a task, and variable-height workstations all extend the working life of experienced employees by years.

The strategic argument is simple and often missed: every year of working life extended is a year of expertise retained. Retaining a sixty-year-old master toolmaker for three extra years — years in which they can also train the next generation — is worth more to most companies than any single automation investment. Design that accommodates aging bodies is not charity. It is workforce planning.

The Perception Problem

The other end of the demographic pipeline is equally broken. Young people are not entering industrial careers in sufficient numbers, and the reasons have as much to do with perception as with pay. Ask a sixteen-year-old to describe a factory and the image is likely straight from the nineteenth century: dark, dirty, dangerous, dead-end. This image is decades out of date, but the industry has done remarkably little to correct it, and the correction is fundamentally a design problem.

The perception problem has three layers, and each requires deliberate design work. The first is the physical layer: many factories genuinely are dark, dirty, and dated. The companies that invest in light, air, color, and clean sightlines — in making the plant a place a young person would be proud to show their friends on Instagram —

report that recruitment changes overnight. The second layer is the digital layer: young workers expect their tools to be as good as their phones. A twenty-year-old who has grown up with intuitive software will not tolerate a control system with a DOS-era interface and a manual thicker than their forearm. The third layer is the narrative layer: industrial work has a story to tell — about building things that matter, about technology, about climate, about making the physical world — and it is not being told. Designers are uniquely equipped to tell it.

Modernizing the workplace is not about making factories look like tech offices, with beanbags and espresso bars. It is about making the work legible and dignifying it: glass walls that show the process, screens that visualize data, tools that feel current, and a physical environment that communicates respect. Every one of these is a design decision, and every one of them is a recruiting argument.

Training and Simulation for Skills

Once young workers arrive, the industry faces its second knowledge problem in miniature: how do you teach a forty-year career's worth of skill in a few years? The answer the best companies have converged on is simulation — and it is one of the great unheralded successes of the digital industrial transition.

Virtual reality training for skilled trades has moved from gimmick to proven pedagogy. Welders can burn through hundreds of simulated joints before touching a real torch, building muscle memory without consuming consumables or risking injury. Assembly workers can practice complex sequences in VR until they are fluent, cutting the learning curve on new products by weeks. Augmented reality overlays guide technicians through unfamiliar repairs with step-by-step visualization, effectively letting a novice perform like an expert on the first attempt. Digital twins of entire production lines allow operators to rehearse changeovers and fault responses in a safe sandbox, turning rare and stressful events into routine practice.

The design insight here is that simulation is not a replacement for the tacit knowledge of the veterans — it is the medium through which that knowledge can be transmitted at scale. When a retiring engineer's expertise is encoded into a simulation scenario, it becomes teachable to an entire cohort, not just one lucky apprentice. The veterans remain the authors of the training; the simulation is their amplified voice. The companies that understand this are not just training workers. They are converting their most irreplaceable asset — accumulated expertise — into a durable, transferable form.

The Dignity of Work

Underneath all of this lies a design value that is easy to dismiss and impossible to avoid: dignity. Industrial work has been systematically devalued in Western culture for a generation, and the consequences are now measurable in labor shortages, skills gaps, and shuttered plants. Design decisions either compound or counter that devaluation.

Consider the difference between two ways of designing the same automation. In the first, the machine is designed to replace the worker, and the worker is reduced to a monitor of a system they do not understand and cannot influence. In the second, the machine is designed to amplify the worker — to take the heavy, repetitive, dangerous parts of the job while leaving the judgment, the craft, and the pride intact. The first produces resentment, turnover, and silent resistance. The second produces the engaged, inventive workforce that every manufacturer claims to want. The technology can be identical. The design intent is everything.

Dignity also means involving workers in the design process itself. The plants that have achieved genuine transformation did not hand down new systems from the engineering office; they put operators in the room where interfaces, workflows, and automations were designed. The workers knew where the friction was, because they lived in it. Their suggestions — a button moved six inches, an alert reworded, a sequence reordered — were often the difference between a system that worked on paper

and one that worked on the floor. Participatory design is not a workshop technique for consultants. In legacy industry, it is the fastest route to both adoption and improvement.

There is an economic argument for dignity, and it is blunt: in a labor market where skilled industrial workers can choose their employer, the companies that treat their people as an afterthought will not get them. The design of the workplace — physical, digital, and cultural — is now a competitive factor as real as price or lead time. The companies that win the next industrial age will be those that understand a simple truth: the worker of the future is not a problem to be automated away, but the scarce resource around which everything else must be designed.

Toolkit Summary

- Treat retiring expertise as a design problem: capture tacit knowledge through sensors, simulation, and co-designed transfer, not just documentation.
- Design for aging bodies as an engineering parameter: ergonomic retrofits, exoskeletons, vision assistance, and assistive automation extend both working life and knowledge retention.

- Recruitment is a design problem: light, clarity, current digital tools, and a dignified physical environment are the most effective workforce marketing a plant can do.
- Use simulation and digital twins to convert veterans' expertise into teachable, scalable training — the veterans remain the authors, the simulation is their amplified voice.
- Design automation to amplify workers, not replace them, and involve operators in design decisions; dignity is the difference between resistance and invention.
- In a scarce labor market, the workplace experience is a competitive factor as decisive as price or lead time.

Chapter 8: Resilience, Reshoring, and Regional Production

IN THE FIRST MONTHS OF 2020, A VENTILATOR MANUFACTURER IN THE AMERICAN MIDWEST DISCOVERED THAT A SINGLE CRITICAL VALVE — A COMPONENT SO SMALL IT COST LESS THAN TWO DOLLARS — WAS MADE IN EXACTLY ONE FACTORY IN THE WORLD, SEVEN THOUSAND MILES AWAY, AND THAT FACTORY HAD JUST BEEN CLOSED BY A PANDEMIC. THE COMPANY COULD NOT MAKE VENTILATORS WITHOUT THE VALVE, AND HOSPITALS COULD NOT SAVE PATIENTS WITHOUT THE VENTILATORS. THE VALVE WAS REDESIGNED IN A WEEK, AND A LOCAL MACHINE SHOP WAS PRODUCING IT WITHIN TEN DAYS. THE CRISIS PASSED, BUT THE LESSON DID NOT: FOR THIRTY YEARS, WESTERN INDUSTRY HAD OPTIMIZED ITS SUPPLY CHAINS FOR A SINGLE VARIABLE — COST — AND HAD QUIETLY OUTSOURCED ITS RESILIENCE ALONG WITH ITS MANUFACTURING. THE PANDEMIC DID NOT CREATE THE FRAGILITY. IT REVEALED IT.

That revelation has changed the design conversation permanently. Supply chain resilience is no longer a logistics topic confined to procurement departments. It has become a design criterion that shapes where products are made, how they are configured, and what they are made of. The next industrial age will be defined as much by the geography of production as by the

technology of production, and designers are now, whether they asked for the job or not, the people who reconcile the two.

The Pandemic Wake-Up Call

The COVID-19 pandemic was a stress test that most global supply chains failed. Personal protective equipment vanished from markets that depended on a single exporting nation. Semiconductor shortages idled automobile plants on three continents for months, because a five-dollar chip was unavailable. Containers piled up in the wrong ports; freight rates multiplied by ten; lead times that had been measured in weeks stretched into quarters. The common thread was not bad luck but design: decades of just-in-time optimization had produced chains that were extremely efficient at rest and catastrophically fragile under stress. They had been engineered for the normal, and the normal had ended.

The economic consequences were severe enough to change policy. The European Union, the United States, and a dozen national governments concluded simultaneously that certain capabilities — semiconductors, pharmaceuticals, critical minerals, medical supplies, defense components — were strategic and could not be left to the market alone. The result has been the most active period of industrial policy in the

West since the 1970s: subsidies, tax credits, procurement preferences, and export controls, all designed to pull production back toward home and allied shores.

For designers, the wake-up call has a specific meaning. Every product designed today will be manufactured into a supply chain that is being redesigned in real time. The location of production, the availability of components, the resilience of logistics — these are now design inputs, as real as material properties or manufacturing tolerances. A design that cannot be produced from available sources in a crisis is a design that fails its primary purpose.

The New Geography of Production

The reshoring and nearshoring trends are real, and they are broad. The United States has seen manufacturing construction spending reach record levels, driven by semiconductor fabs, battery plants, and electric vehicle assembly — investments that would have been unthinkable offshore a decade ago. Mexico has emerged as the great beneficiary of nearshoring, absorbing manufacturing capacity that once went to Asia, thanks to proximity, trade agreements, and the logic of shorter supply lines. In Europe, production is regionalizing along similar lines, with Eastern European nations like Poland, Romania, and the Czech Republic attracting investment as Western European companies shorten their chains. The pattern is not the wholesale return of manufacturing to the West — the era of cheap offshore labor for mass

production is not coming back either. It is a rebalancing: more production closer to consumption, more redundancy, more strategic intent.

This new geography changes the design brief. Products designed for a global supply chain assumed a certain kind of scale: one factory, one process, one set of tooling, amortized over worldwide volume. Regional production implies a different model — multiple smaller plants, each serving a regional market, each with local variations in skills, materials, and regulations. Designing for that model is a distinct discipline, and it is largely new.

Designing for Regional Production

The first principle of design for regional production is local sourcing. When a company commits to making a product in a region, the design must be able to use what the region can supply. This is a reversal of the old logic. For decades, designers specified materials and components first and asked where they came from later — or never. The resilient approach starts with the question: what can be sourced within a radius of five hundred kilometers? Steel, aluminum, castings, plastics, electronics, fasteners — every material decision becomes a sourcing decision, and the design must be built around the answer.

This constraint is not purely a burden. Local sourcing compresses lead times, reduces shipping costs and carbon footprints, and insulates the product from

currency swings and trade disputes. It also creates a design advantage: products that can be made from regional materials and regional skills can be adapted to regional markets without redesigning from scratch. The trade-off is real — some materials and components simply are not available locally at the same cost or quality — but the starting assumption must be flipped. The question is no longer "where is the cheapest source?" but "what can we make well, near our customers?"

The second principle is modular design. The great structural answer to the tension between global efficiency and regional resilience is the modular architecture: a product designed as a set of standardized interfaces connecting interchangeable modules. The core — the part that carries the intellectual property, the quality-critical engineering, the expensive tooling — can be produced centrally at scale. The periphery — the housing, the local variants, the service parts — can be produced regionally, close to the customer. Modular design lets a company have both worlds: the cost advantages of scale where they matter, and the resilience of local production where it matters.

The third principle is design for local assembly and repair. A product that can be assembled, disassembled, and repaired by regional workers with regional tools is a product that creates regional jobs and regional resilience simultaneously. Fasteners instead of welds where possible. Standardized connectors instead of proprietary ones. Service documentation designed for the people who

will actually open the machine. These choices — which are pure design choices — determine whether a product can be sustained by a local economy or becomes a captive of its manufacturer.

There is a fourth principle, less discussed but equally important: design for the skills that exist. A region with a strong welding culture but a thin electronics base can produce a machine that is welded locally and fitted with an imported electronic core. A region with skilled machinists but no injection molding capacity can make the same product with machined parts and a molded component shipped in. The design that respects regional capability — that asks what the local workforce can actually do, and shapes the product around that — converts the constraint of location into a source of competitive advantage. It also compounds over time: every product designed around local skills trains and deepens those skills, so the region's capability grows with each generation of design. This is how the industrial commons — the ecosystem of skills, suppliers, and knowledge that makes a place good at making things — is rebuilt, one design decision at a time.

The Trade-Offs: Cost Versus Resilience

None of this is free, and the honest conversation about reshoring must begin with the arithmetic. Offshore production of labor-intensive goods can still be 20 to 40 percent cheaper on unit cost, and no amount of patriotic

rhetoric changes that. Resilience is a form of insurance, and insurance has a premium. The companies that are making the reshoring decision work are those that treat the premium as a design problem rather than a political slogan — finding ways to reduce the cost penalty through automation, through design for manufacturability, through products that are deliberately engineered to be made by machines rather than by cheap hands.

The newer insight, though, is that the true cost comparison is not between offshore unit cost and onshore unit cost. It is between the total cost of a fragile system and the total cost of a resilient one. The semiconductor shortage of 2021–2023 cost the global automotive industry more in lost production — estimates ran to hundreds of billions of dollars — than any plausible reshoring premium. A single missed shipment at a critical moment can erase a decade of procurement savings. Once risk is priced in, resilience is frequently the cheaper option, and design is the tool that prices it in: designing products that can be sourced from multiple suppliers, that can switch between materials, that can be produced at different volumes in different places. Flexibility is the design expression of resilience, and it is purchasable.

Policy as a Design Input

The new industrial policy is a fact of life for designers, and it cuts both ways. The CHIPS and Science Act in the United States has committed tens of billions of dollars to

semiconductor production and research, with the explicit goal of building resilient domestic supply. The European Union's response — the European Chips Act, the Net-Zero Industry Act, the Critical Raw Materials Act — pursues the same logic: strategic autonomy, domestic capacity, reduced dependence. Tax credits for clean technology, procurement preferences for domestic suppliers, and export controls on advanced technologies all shape the design space. A designer who ignores them does so at their peril; a designer who understands them can build on them.

But policy also creates obligations that design must meet. Subsidized factories are not blank checks; they come with conditions on labor standards, environmental performance, and community benefit. The design of the products that emerge from these factories — and of the factories themselves — will be scrutinized as never before. The industrial age that policy is trying to rebuild is not the industrial age of the twentieth century, with its smoke and its scars. It is an industrial age that must be clean, safe, and visibly beneficial to the regions that host it. That is a design brief, whether the politicians who wrote the laws knew it or not.

Designing Supply Chains, Not Just Products

The deepest shift in thinking is the one hardest to see: the supply chain itself is now a designed artifact. For most of the industrial era, companies designed products and then bought a supply chain to make them. The resilient companies of the next age reverse the order. They design the supply chain — its geography, its redundancy, its flexibility, its relationships — and the product is designed to fit it.

This is visible in the best practice of the moment. Automotive companies are co-locating battery and motor production with assembly plants, designing vehicles around regional cell supply. Aerospace companies are building multi-source strategies for forgings and castings into their design reviews. Electronics manufacturers are designing chips and boards with second-sourcing built in from the start, not bolted on after a crisis. In every case, the design team's questions have expanded: not just "how will this be made?" but "where will this be made, by whom, with what, and what happens if that answer changes?" Those questions are now as central to industrial design as the shape of the product itself.

The pandemic delivered a brutal lesson in the cost of designing for efficiency alone. The next industrial age will be built on a different value: designing for the ability to keep making things, no matter what the world does. That is not a supply chain strategy. It is a design philosophy.

Toolkit Summary

- Treat resilience as a design criterion: source locally where possible, multi-source critical components, and design products that can switch suppliers and materials under stress.
- Regional production changes the design brief — build the design around regional materials, skills, and regulations rather than a single global process.
- Modular architecture is the structural answer: standard interfaces let the core be produced at scale while regional modules, assembly, and repair stay close to customers.
- Price risk into cost comparisons: fragility has a measurable cost (lost production, missed shipments) that frequently exceeds the reshoring premium.
- Policy is a design input: CHIPS Act, EU industrial strategy, tax credits, and procurement rules shape what can be built where — understand them or be constrained by them.
- Design the supply chain, not just the product: geography, redundancy, and flexibility are now first-order design decisions.

Chapter 9: The Aesthetics of Heavy Industry

THE EXCAVATOR SITS ON THE DEALER'S LOT LIKE A PROMISE. ITS PAINT IS A YELLOW THAT HAS BECOME SO ICONIC IT HAS A LEGAL EXISTENCE — A TRADEMARKED SHADE, PROTECTED IN DOZENS OF JURISDICTIONS, RECOGNIZED BY CONSTRUCTION WORKERS AND CHILDREN ALIKE. ITS SILHOUETTE IS UNMISTAKABLE AT A THOUSAND METERS. ITS CAB IS GLAZED LIKE A PIECE OF MODERN ARCHITECTURE. NOBODY BUYS AN EXCAVATOR BECAUSE IT IS BEAUTIFUL. YET THE EXCAVATOR'S MANUFACTURER SPENDS REAL MONEY EVERY YEAR MAKING SURE IT IS. THIS IS NOT VANITY. IT IS THE MOST MISUNDERSTOOD COMPETITIVE WEAPON IN INDUSTRIAL BUSINESS.

For most of the twentieth century, heavy industry treated aesthetics as an afterthought, if it treated them at all. Machines were engineered, not designed; paint was applied to prevent rust, not to communicate; the buyer was an engineer who cared about specifications, and specifications were believed to be immune to beauty. That era is over. The most successful industrial companies of the present — the ones that command premium prices, attract the best workers, and endure across generations — treat appearance, identity, and emotion as engineering

disciplines. The aesthetics of heavy industry has become a serious business, and it is reshaping what industrial products are and what they mean.

Why Aesthetics Matter for Industrial Products

The argument against aesthetics in industrial design has always been the same: machines are bought for what they do, not how they look. The argument has a core of truth and a shell of error. It is true that an excavator that breaks down will not be saved by a beautiful paint job. It is false that beauty has no bearing on performance — because the people who operate, maintain, and buy machines are human, and humans are moved by what they see.

Start with the operator. A machine that is pleasant to look at, whose controls feel considered, whose cab is a well-designed place to spend a twelve-hour shift, is a machine that is operated better. Aesthetics here merges with ergonomics: the visual clarity of a control layout, the legibility of a display, the calm of a well-proportioned cab — these reduce fatigue and error as surely as any technical specification. Then consider the owner. A machine is a capital asset that represents a business's competence to its own customers. A contractor who shows up to a job with clean, modern, well-designed equipment is signaling reliability before a single word is spoken. The machine is a uniform, a business card, and a

portfolio piece at once. Finally, consider the brand. In an industry where products from different manufacturers are functionally near-identical, appearance is often the only differentiator a customer can articulate. Aesthetics is how a company says who it is.

There is also a subtler economic argument. The cost of good industrial aesthetics is small — a fraction of a percent of product cost in most cases, spent on industrial designers, color and materials specialists, and brand discipline. The payoff accrues across the entire business: higher perceived quality, pricing power, operator preference, resale value, and workforce pride. Few investments in industrial business have a better ratio of cost to return.

The aesthetics of sustainability is accelerating this shift. As industries electrify and decarbonize, the visual language of machines is changing with them — and the change is a design opportunity. An electric excavator has no exhaust stack, no diesel tank, no reason to be painted the color of soot and grease; its designers can make it look as clean as its powertrain. The manufacturers that have led electrification are visibly using this: new silhouettes, new colorways, quieter presence — machines that communicate a new era before the operator even climbs aboard. The aesthetic here is not decoration; it is truth. A machine that looks like its future, and not like its grandfather, tells customers, regulators, and workers that

the company has actually changed. In an industry where perception lags reality by decades, the fastest way to close the gap is design.

From Function to Brand

The shift from purely functional to brand-led industrial design happened in plain sight, though it is usually told as a consumer story. The same decades that turned the smartphone into an object of desire transformed industrial machinery. The pioneers were the Japanese manufacturers who entered Western markets in the 1970s and 1980s — Komatsu, for example, built a global brand on the back of equipment whose quality was visible in its finish, and whose identity was consistent across every product it made. Western incumbents, startled out of complacency, began hiring industrial designers in earnest.

The result is a design language that is now as deliberate in heavy industry as in any consumer category. A brand's machines are recognizable at a glance, sharing proportion, color, and detailing across product lines. Cab interiors are designed as coherent environments, with the same attention to sightlines, reach, and display hierarchy as a premium automobile. Even the sound of a machine — the note of an engine, the acoustics of a cab — is now engineered as part of the experience. This is brand thinking applied to the largest objects most people will ever see up close, and it has transformed the industry's

self-conception. Machine manufacturers no longer think of themselves as selling iron. They think of themselves as selling identity, capability, and trust, in iron form.

The Designers of Yellow Iron

Three companies illustrate how far this transformation has gone, and each does it differently. Caterpillar's yellow is perhaps the most valuable industrial color in the world — a shade so associated with the brand that the company guards it jealously and deploys it with military consistency. But Cat's design discipline goes far beyond paint. Its machines share a family resemblance that lets a dealer's lot read as a single coherent statement; its cab interiors are designed around operator comfort as a productivity argument; and its brand extends into clothing, toys, and footwear — proof that industrial aesthetics can generate consumer emotion strong enough to walk off the job site.

Volvo has built its industrial identity on a different pillar: safety, expressed visually. Volvo construction equipment and trucks share the design DNA of the car division — robust, human-centered, Scandinavian in their clarity — and the brand's entire aesthetic vocabulary whispers protection and control. The result is that Volvo machines are bought partly for what they say about the buyer: that safety and people matter. That is an emotional purchase in an industry that claims to make only rational ones.

John Deere has done something stranger and more impressive: it has built brand loyalty normally reserved for sports teams. The green and yellow livery, unchanged in its essentials for over a century, is worn by farmers as clothing and flown as flags. Deere's design discipline extends to the dealer experience, the parts system, and the digital interfaces that now guide the machines — the brand is the product, and the product is the brand, in a loop that has made Deere one of the most valuable agricultural companies on earth. What these three prove is that there is no single formula for industrial aesthetics. There is only the discipline of choosing an identity and expressing it without compromise.

The Emotional Connection to Industrial Objects

It is easy to underestimate the emotional weight of industrial objects, and harder to do so after spending time with the people who use them. Ask a farmer about a tractor that served the family for thirty years and the word "love" appears without embarrassment. Ask a miner about the haul truck they drove for two decades, and you will hear a eulogy. These machines are not tools; they are partners in a working life. They are present at the hardest moments of the year, reliable in conditions that would defeat a person, and visibly, physically real in a world that grows more virtual by the day.

Designers who understand this relationship design differently. They give machines names and personalities. They craft details — a handhold placed exactly where a gloved hand reaches, a sightline that lets the operator see the bucket teeth touch the ground — that tell the operator the machine was designed for them, by people who understood the work. These are the details that create the difference between a machine that is used and a machine that is loved, and the loved machine is maintained better, lasts longer, and is replaced with the same brand.

This emotional connection is also generational and cultural. For communities built around industry — mining towns, shipbuilding cities, agricultural regions — the machine is part of identity. The design of industrial products therefore carries a social responsibility that is rarely acknowledged: a well-designed machine honors the people who use it, and a cynically designed one demeans them. The dignity of the work, and of the worker, is expressed in the object.

Designing for Pride

Pride is the most underrated variable in industrial economics. Workers who are proud of their machines and their workplace produce better work, stay longer, and recruit their friends. Owners who are proud of their equipment invest in it, maintain it, and defend the brand in the marketplace. Communities that are proud of their

industry defend it politically and welcome its expansion. Pride is not a soft factor. It is a hard asset, and it is manufactured, in large part, by design.

The mechanics of pride are straightforward. Pride requires that the object be worthy of respect: well-proportioned, well-finished, honest in its materials, thoughtful in its details. It requires that the brand be consistent, so that the object stands for something larger than itself. And it requires that the people who make and use the object be able to see themselves in it — that the design reflects the values of the work, not the whims of a stylist. When all three conditions are met, the machine becomes what the best industrial design has always been: a source of identity for everyone it touches.

Awards and the Competitive Advantage of Beauty

The rise of industrial design awards — the Red Dot, the iF Design Award, the International Design Excellence Awards, the Design Council's awards in the UK — used to be a consumer phenomenon. Today the winners' lists are crowded with excavators, agricultural machinery, medical devices, and industrial equipment, and the awards have become a serious marketing instrument for B2B manufacturers. Winning is celebrated internally, displayed in dealer showrooms, and used to justify premium pricing. The awards matter because they certify something that buyers cannot easily verify for

themselves: that a manufacturer takes design seriously, and that its products embody a standard beyond mere function.

The deeper significance is competitive. As industrial products converge in performance — everyone's excavator digs, everyone's tractor plows, everyone's turbine generates — the battleground shifts to experience, identity, and emotion. Companies that treat design as a cost center are competing on a single variable, price, in an industry where the lowest-cost producer is usually someone else. Companies that treat design as a strategic function compete on meaning, and meaning is harder to copy than a specification. In the next industrial age, the machines that win will not just work better. They will look, feel, and mean something better. The companies that understand this are already pulling ahead, one yellow silhouette at a time.

Toolkit Summary

- Aesthetics is an engineering discipline in heavy industry: operator experience, perceived quality, and pricing power all respond to design investment.
- Brand-led design means a consistent identity — proportion, color, detailing — across every product line, from cab interiors to dealer lots.

- Study the masters: Caterpillar (consistency and iconic color), Volvo (safety expressed visually), John Deere (a century-old livery that generates genuine loyalty).
- Machines are emotional objects — design details that honor the operator's work create the difference between a machine used and a machine loved.
- Pride is a hard asset: workers, owners, and communities who are proud of the machines produce measurable economic returns.
- Use awards and design recognition as competitive instruments: they certify design seriousness to buyers who cannot verify it themselves.

Chapter 10: Case Studies in Reinvention

EVERY LEGACY INDUSTRIAL COMPANY FACES THE SAME QUESTION: HOW DO YOU KEEP THE STRENGTHS THAT BUILT YOU — THE ENGINEERING DEPTH, THE CUSTOMER TRUST, THE INSTALLED BASE — WHILE BECOMING SOMETHING THE NEXT DECADE DEMANDS? THE ANSWERS ARE AS VARIED AS THE COMPANIES THAT GIVE THEM, AND THEY ARE WORTH STUDYING IN DETAIL, BECAUSE REINVENTION IS NOT A SLOGAN. IT IS A SEQUENCE OF SPECIFIC DECISIONS ABOUT PRODUCTS, SERVICES, AND IDENTITY, MADE UNDER PRESSURE, WITH REAL MONEY AT STAKE.

The seven companies examined here — Rolls-Royce, GE, Volvo, Siemens, Caterpillar, Airbus, and John Deere — represent the full spectrum of industrial reinvention. Together they demonstrate every major strategy available to a legacy manufacturer: selling outcomes instead of products, digitizing the core, leading with safety and sustainability, building platforms for automation, redesigning production itself, and embedding intelligence in the field. Each got something right, each got something wrong, and each left lessons that apply far beyond its own industry.

Rolls-Royce: Selling Outcomes, Not Engines

The story of Rolls-Royce's reinvention begins with a crisis of its own making. In the 1960s, the company's gamble on a new jet engine bankrupted it; the British government nationalized it, and the company emerged leaner, focused on aerospace and marine power. The modern reinvention began with a realization: airlines do not want to own engines, they want to fly. From that realization came "power by the hour," the servitization model that made Rolls-Royce the most studied example of its kind. Under the model, airlines pay for engine availability — for hours flown, for thrust delivered — while Rolls-Royce retains ownership of the hardware and is responsible for its maintenance, reliability, and performance. The customer buys an outcome; the manufacturer bears the risk and reaps the reward of engineering excellence.

The design implications were profound. When you sell hours of operation rather than iron, every design decision is measured against lifetime cost and reliability, not just first cost. Engines are designed for maintainability, with modular cores that can be swapped on the wing. Sensors and telemetry stream performance data continuously, so maintenance is predictive rather than scheduled. The engine becomes a service platform, and the manufacturer's relationship with the customer runs for decades rather than months.

What went right: the model aligned incentives perfectly, created a durable revenue stream, and generated a data advantage that competitors could not match. What went wrong: the model concentrates risk. When a single engine family — the Trent 1000 — suffered from premature turbine blade corrosion, Rolls-Royce bore the entire cost of grounding and repair, and its profits collapsed for years. Servitization is a brilliant design if you can control the variables; it is brutal if you cannot. The lesson for every legacy manufacturer is that selling outcomes is the highest-value form of industrial design — and the highest-risk one. The mitigation is design rigor itself: reliability, maintainability, and data, engineered in from the start.

GE: The Cautionary Tale of the Digital Pivot

General Electric was the original industrial conglomerate, and its attempt to become a "digital industrial" company is the most instructive failure of the era. In 2015, GE announced that it would become a top-ten software company by 2020, built around Predix, its industrial internet-of-things platform. The logic was seductive: GE's machines generated oceans of data, and the company that owned the data platform would own the industrial future. Billions were invested, thousands of software engineers were hired, and the narrative was repeated so often that GE's own leadership came to believe it.

The execution collapsed. Predix was sold to customers as a platform before it was a product; the technology was immature, the go-to-market confused, and the internal politics ferocious. GE's industrial businesses, starved of management attention, deteriorated. The company that had once defined American manufacturing ended the decade dismantled, with the digital business sold off at a fraction of its invested cost. The failure was not a failure of vision — industrial software is a real and growing market, and GE was early to it. It was a failure of design in the broadest sense: the transformation was designed as a financial narrative for investors rather than as a working system for customers, and the organization was never designed to run a software business alongside a hardware one.

The lessons are worth their weight in gold. Digitization is not a strategy; it is a capability, and it must be built where it can be tested against real customers, not announced to the world. A legacy industrial company can become a software company, but only by redesigning its organization, its talent model, and its incentives around software — which is a design problem of the highest order. GE's tragedy is that it had the right insight and the wrong process. The insight survived; the company did not.

Volvo: Safety as Identity, Electrification as Future

Volvo has spent decades proving that a legacy manufacturer can build its entire identity on a value rather than a product. Safety is not a feature of Volvo's design; it is the reason the company exists, expressed in every platform decision, every crash-test investment, and every public statement. The value has survived changes of ownership, market cycles, and the shift from Swedish independence to Chinese ownership under Geely — because it is a design principle, not a business unit. When Volvo says it designs for people, it means it in a way that customers can feel.

The reinvention currently underway extends the same logic to electrification and sustainability. Volvo has committed to becoming an electric-only manufacturer, has embedded sustainability targets into every product program, and is redesigning its cars, trucks, and construction equipment around the electric drivetrain. The design challenge is formidable: heavy equipment does not electrify as easily as cars, and the infrastructure, battery supply, and charging ecosystems are not yet in place. But Volvo's approach shows the correct pattern for a legacy company: take the value that defines you — safety, in Volvo's case, extended to the safety of the climate — and redesign the entire portfolio around it, decade by decade.

What Volvo got right is coherence: the brand, the products, the technology roadmap, and the public commitments all point the same direction, and customers trust the direction because the company has been consistent for sixty years. What remains unproven is whether the premium that consistency commands will survive the brutal economics of electrification. The lesson is that identity is a strategic asset with a long half-life — and that the companies that define themselves by values, not products, have the freedom to change everything about what they make.

Siemens: The Digital Enterprise

Siemens took a different route to the same destination as GE: instead of announcing a separate digital business, it made digitalization the operating system of its existing ones. The "Digital Enterprise" strategy integrated design, simulation, and manufacturing software across the company's portfolio — from the product lifecycle management tools that design machines to the automation systems that run factories to the digital twins that model both. The result is a company where the boundary between physical and digital has been deliberately dissolved, and where customers are sold not machines but integrated capabilities.

The success of Siemens's approach is visible in its resilience. Its digital industries division has become the profit engine of the group, growing through cycles that

punished its conventional automation competitors. Its factory in Amberg, Germany — which operates with an automation rate above seventy-five percent and a defect rate measured in parts per million — is the most cited proof that a legacy manufacturer can turn its own production into both a demonstration and a product. What Siemens got right was sequencing: it digitized itself before selling digitalization to others, so the sales pitch was evidence rather than promise. What it got wrong, in the eyes of many observers, is the same thing: the transformation was so deeply integrated that it is hard to point to a single moment of reinvention — the company changed everything by changing everything, which is a hard act for most organizations to copy. The lesson: integration is a strategy, but it is a strategy that requires a long runway and a leadership team willing to make the whole company the product.

Caterpillar: Autonomy in the Pit

Caterpillar's reinvention is happening where its machines work: in the mines. Autonomous haulage — driverless trucks moving ore around open-pit mines — was pioneered by Caterpillar in the 1990s and has quietly become one of the largest commercial deployments of autonomy in the world. Thousands of autonomous trucks now operate around the clock in mines from Australia to Chile, moving more material per hour than their manned counterparts, with fewer accidents and lower operating

costs. The design achievement is not the self-driving technology alone; it is the integration of autonomy into a working system — dispatch software, fleet management, maintenance schedules, and safety protocols — that operators trust with billions of dollars of equipment.

What Caterpillar got right is incrementalism with a clear end state. It did not announce the autonomous mine and deliver it in one leap; it introduced autonomy where the economics and the safety case were strongest, proved the model, and expanded. It also understood that autonomy does not remove the human — it changes the human's role. The miners moved from the cab to the control room, and Caterpillar designed that transition as carefully as the trucks themselves, with training, interfaces, and career paths for the people whose jobs changed. What it got wrong is the pace of the broader transformation: the company was slow to embrace the software and services businesses around its machines, leaving room for nimbler competitors and for Caterpillar's own later scramble into connectivity and subscription models. The lesson: the hard part of automation is not the machine. It is the system around the machine, including the humans, and the companies that design all of it together are the ones that win.

Airbus: The Factory as a Product

Airbus has spent the last decade proving that a legacy manufacturer can reinvent how it makes things, not just what it makes. Its "Factory of the Future" program has transformed final assembly — the most conservative corner of aerospace — with automated drilling and fastening, collaborative robots, exoskeletons for workers, and a digital thread that tracks every part from raw material to flight. The A350 program, designed from the start around digital manufacturing, cut assembly times dramatically and demonstrated that a complex, safety-critical product could be produced in a genuinely digital factory.

The design significance of Airbus's work is that it treats production itself as a designed system rather than an inherited condition. Layout, logistics, automation, and human roles are designed together, with the worker as a first-class component: robots handle the heavy, repetitive, hazardous work, while humans do the judgment, the inspection, and the unexpected. The company has also used its production capability as a strategic weapon, scaling output to levels that its competitors have struggled to match. What Airbus got right is the willingness to rebuild its own factory floor as a demonstration of its own future — the factory is the proof. What it got wrong is the cost: the transformation has consumed enormous capital and management attention, and the company's production crises — engine delays, supplier failures — show that a digital factory still

depends on a physical supply chain that cannot be redesigned overnight. The lesson: a factory of the future is a product like any other, and it needs the same design rigor, the same user research, and the same honest accounting as the aircraft it builds.

John Deere: Intelligence in the Field

John Deere's reinvention answers a different question: what happens when the product leaves the factory and lives in a chaotic, uncontrolled environment? Deere has spent two decades embedding intelligence into agricultural machinery — GPS guidance, variable-rate application, yield mapping, and now fully autonomous tractors and implements. The See & Spray system, which uses computer vision to distinguish weeds from crops and apply herbicide only where needed, is a design triumph: it cuts chemical use dramatically, which is good for the environment, the farmer's costs, and Deere's margins. The company has repositioned itself from a maker of tractors to a provider of agricultural outcomes, with software subscriptions, data services, and precision platforms that deepen its relationship with every farmer who uses them.

What John Deere got right is the long game. It invested in precision agriculture for decades before the economics worked, built the brand trust that made farmers willing to share their data, and designed the technology to respect the farmer's expertise rather than replace it — the

machines assist, the farmer decides. What it got wrong, or is still learning, is the tension inherent in the model: farmers who have historically owned their machines and their independence are wary of a future where the software, and the data, belong to the manufacturer. The right-to-repair movement, and the legislative battles around it, are the direct result of this design decision. The lesson: when you put intelligence in the field, you are designing a relationship, not just a machine, and the terms of that relationship — who owns the data, who can repair the system, who benefits from the insights — are design decisions with political consequences.

The Pattern Beneath the Cases

Seven companies, seven strategies, one underlying pattern. Every successful reinvention was anchored in a durable identity — Rolls-Royce's engineering excellence, Volvo's safety, Deere's farmer-first ethos — and every failure, GE's above all, came from abandoning identity for narrative. Every success integrated the human into the new system rather than designing around them. Every success treated reinvention as a design problem with multiple variables — product, service, organization, production, and relationship — and worked them together. The companies that treat reinvention as a single big bet fail; the ones that treat it as a continuous design practice, applied to everything they do, survive and

compound. That is the true lesson of the case studies: in the next industrial age, reinvention is not an event. It is the job.

Toolkit Summary

- Rolls-Royce proves the power and risk of servitization: selling outcomes aligns incentives and builds data advantages, but concentrates risk on the manufacturer — mitigate with reliability engineered in from the start.
- GE is the cautionary tale: digitization announced as a financial narrative, without a working product and an organization designed for software, destroys value even with the right vision.
- Volvo shows that identity is strategic: a value (safety, extended to sustainability) coherently applied across the portfolio survives ownership changes and buys trust for radical transitions.
- Siemens demonstrates integration: digitize yourself before selling digitalization; the factory is the evidence, and integration is a strategy with a long runway.

- Caterpillar's autonomy program shows that automation succeeds when the system around the machine — dispatch, safety, training, and changed human roles — is designed as carefully as the machine itself.
- Airbus treats production as a designed product: the factory of the future is proof, demonstration, and competitive weapon — but it still depends on a physical supply chain.
- John Deere's precision agriculture shows that embedded intelligence designs a relationship, not just a machine: data ownership and right-to-repair are design decisions with political consequences.
- The pattern: anchor reinvention in durable identity, integrate humans into new systems, and treat reinvention as continuous design practice rather than a single big bet.

Chapter 11: The Road Ahead

EVERY CHAPTER OF THIS BOOK HAS POINTED TO THE SAME DESTINATION. LEGACY INDUSTRY IS NOT DYING; IT IS BEING REDESIGNED, AND THE REDESIGN IS NOT A TECHNOLOGY PROJECT, A COST PROGRAM, OR A MARKETING EXERCISE. IT IS A COMPREHENSIVE ACT OF DESIGN — COVERING PRODUCTS, PRODUCTION, SUPPLY CHAINS, WORK, AND MEANING — AND IT IS BEING LED, UNEVENLY, BY COMPANIES THAT UNDERSTAND THIS AND RESISTED BY COMPANIES THAT DO NOT. THE QUESTION THAT REMAINS IS WHAT THE NEXT DECADE LOOKS LIKE FOR THE COMPANIES THAT GET IT, THE DESIGNERS WHO LEAD IT, AND THE SOCIETIES THAT DEPEND ON IT. THIS CLOSING CHAPTER LAYS OUT THE SYNTHESIS: THE SHAPE OF THE INTEGRATED DESIGN ORGANIZATION, THE CASE FOR DESIGN LEADERSHIP AT THE TOP, THE SKILLS THE NEXT GENERATION OF DESIGNERS WILL NEED, THE ROLE OF POLICY AND EDUCATION, AND A FIVE-YEAR OUTLOOK THAT SEPARATES WHAT WILL CHANGE FROM WHAT WILL NOT.

The Integrated Design Organization

For most of the industrial era, design lived in a box. In consumer companies, the box was called industrial design or styling, and it sat at the end of the product

development process, adding surface to machines that were otherwise finished. In industrial companies, the box often did not exist at all — engineering absorbed every decision, and the word "design" meant the drawing office. Both arrangements are obsolete, and the organization that replaces them is the single most important structural change of the next industrial age.

The integrated design organization treats design as a function that spans the whole business. It is present at the very beginning, when strategy is set and the question is not "what will we make?" but "what will we mean?" It is present in product architecture, where the modularity that enables regional production and the serviceability that enables outcome-based business models are decided. It is present in production design, where the factory is designed as deliberately as the product. It is present in the digital layer, where interfaces, data, and algorithms are designed with the same human-centered rigor as physical objects. And it is present in the field, where service, training, and the long-term relationship with the customer are designed as part of the offer.

The distinguishing mark of the integrated design organization is not a bigger design department. It is that design decisions are made by people whose job is to see the whole system, at every level of the company. This requires something most legacy companies do not yet have: a design function with authority, and designers whose skills extend beyond the aesthetic into the strategic.

Design Leadership at the Board

The governance consequence follows directly. If design is a strategic function, it needs strategic leadership — a voice at the board table, with the same standing as engineering, finance, and operations. The companies that have appointed chief design officers and given them genuine authority are still a minority, but the pattern is unmistakable: in the companies where design has board-level sponsorship, transformation moves faster, product decisions are more coherent, and the brand compounds rather than fragments.

The case for design leadership is not aesthetic. It is that design is the only function whose native question is "what is this for, and for whom?" In an era of compressed product lifecycles, converging technology, and existential pressure to change, that question is the most strategically valuable one a company can ask. The CFO can price the future, the CTO can build it, and the COO can run it — but someone must decide what the future is for, and that is the designer's question. Boards that do not have a credible answer to it are flying blind.

This is also a pipeline problem. Most board-level designers of the next decade are working today as senior designers, design directors, or heads of innovation, and most of them are not being developed for leadership. The companies that identify design talent with strategic

instincts, rotate it through business roles, and prepare it for the board will have a structural advantage that competitors cannot copy quickly.

The Skills Designers Need Next

The designer of the next industrial age is not the designer of the last one, and the gap between the two is where careers and companies will be made. The traditional industrial designer's toolkit — sketching, modeling, materials, manufacturing processes, human factors — remains necessary, but it is no longer sufficient. Five additional capabilities separate the designers who will lead from those who will be bypassed.

The first is systems thinking. The next industrial age is not designed object by object but system by system: product, production, supply chain, service, and data as one continuous design problem. Designers must be able to hold the whole system in view, to see how a decision about a material affects a factory in another country and a service model a decade away. The second is data literacy. Industrial design increasingly operates on streams of sensor data — performance, failure, usage — and the designer who cannot read, question, and shape data will design in the dark. This does not mean becoming a data scientist; it means being able to hold a conversation with one, and to translate insight into form and behavior. The third is the digital-physical craft: designing interfaces, algorithms, and autonomy as

confidently as designing steel and plastic, and understanding where each belongs. The fourth is economic and business fluency. The designer who can build a business case, read a P&L, and defend a design decision in the language of the boardroom will win budget wars that the purely aesthetic designer loses. The fifth is the hardest to teach: judgment about people. The design of work, the dignity of workers, the emotional relationship between humans and machines — these are the core material of the next industrial age, and they require designers who genuinely understand how people think, feel, and work, and who care.

The Role of Policy and Education

No company reinvents itself in a vacuum, and the next industrial age will be shaped by two institutions that currently lag behind the companies they serve: policy and education. The industrial policy wave of the current decade — chips acts, net-zero industry acts, reshoring incentives — is creating the demand for design capability whether or not its authors intended it. The factories being subsidized must be designed; the products emerging from them must be designed; the workers staffing them must be trained and retained. The opportunity for policymakers is to make design capability an explicit part of industrial strategy, in the way that Germany's Fraunhofer Institutes and the UK's Catapult network already do: public infrastructure that connects research, design, and

industry. The risk is that the money flows to concrete and machines while the design capability — the thing that determines whether the machines are worth building — is left to chance.

Education is the longer lever. The industrial design curricula of most universities still reflect a consumer-products world, graduating designers who can make a beautiful chair and cannot read a factory. The schools that will matter in the next decade are the ones teaching design and engineering as a single discipline, with manufacturing, supply chains, and industrial policy in the curriculum, and with students spending real time on factory floors. The apprentice model — the way industrial expertise has always actually been transferred — deserves revival in design as much as in the trades: design education that combines academic depth with years of supervised practice in real industrial settings. The countries and companies that invest in this pipeline will reap the benefits for decades; the ones that wait will import the talent or do without it.

Five-Year Outlook: What Will Change, What Will Not

The next five years are close enough to predict with confidence, and the predictions divide cleanly into what will change and what will not. What will change: design will move up the industrial agenda. The number of industrial companies with board-level design leadership

will grow significantly, and design will be a named component of more transformation programs. Sustainability and resilience — the two forces that have done more than any marketing campaign to elevate design — will keep pulling design into the center of strategy, because both are fundamentally design problems. The digital layer of industrial products will keep expanding, and with it the demand for designers who can shape software, data, and autonomy. Regional production will continue its uneven rebalancing, and with it the design practices of local sourcing and modular architecture will become standard rather than exceptional.

What will not change: the fundamentals of industrial value. Physics will still matter — materials, tolerances, energy, and wear will still decide which machines earn their keep. The human will still matter — the aging workforce will still be the repository of expertise, and the young workforce will still be the scarce resource, whatever the technology does. Trust will still matter more than anything — the companies that keep their promises to workers, customers, and communities will keep their position, and the ones that don't will lose it regardless of their design sophistication. And the work itself — the dignity of making real things, well — will still be the emotional core that design serves. These fundamentals are why this book could be written at all: the next industrial age will be built on the same human values as the last one, with new tools.

The Design Principles for the Next Industrial Age

What remains is to state plainly what the preceding chapters have argued implicitly. The design principles for the next industrial age can be counted on two hands, and every company that survives the transition will honor most of them.

First, design the whole system, not just the product. Second, design for the worker as the scarce resource: dignity, safety, and expertise are design inputs, not afterthoughts. Third, design for resilience as seriously as for efficiency — geography, redundancy, and flexibility are as real as cost. Fourth, design for a finite planet: sustainability is not a constraint on design but the context in which all design now happens. Fifth, design for the emotional relationship between people and machines — pride is an asset, and it is manufactured. Sixth, anchor every reinvention in a durable identity; the companies that know what they mean survive what they make. Seventh, treat the factory as a product, the supply chain as a design, and the customer relationship as a service. Eighth, put data in service of human judgment, never in place of it. Ninth, bring design to the boardroom, because the question "what is this for, and for whom?" is the most strategic question there is. And tenth, remember that the industrial age is not ending — it is beginning again, and it

will be designed, deliberately, by the people who understand that making things is one of the most meaningful things humans do.

The road ahead is not a prediction. It is a design brief.

Toolkit Summary

- Design is becoming a strategic function spanning product, production, supply chain, and service — the integrated design organization replaces the design department.
- Board-level design leadership is the governance consequence: the question "what is this for, and for whom?" is the most strategically valuable a company can ask.
- The next designer needs five capabilities beyond the traditional toolkit: systems thinking, data literacy, digital-physical craft, business fluency, and judgment about people.
- Policy must fund design capability, not just concrete and machines; education must teach design and engineering as one discipline, with real factory-floor practice.
- Five-year outlook: design moves up the agenda, sustainability and resilience keep pulling it to the center, and the digital layer expands — while physics, the human, and trust remain the unchanged fundamentals.

- Ten design principles for the next industrial age: design the whole system; honor the worker; build resilience; design for a finite planet; engineer pride; anchor in identity; treat factory, supply chain, and service as designed; put data in service of judgment; lead with design at the board; and remember that making things well is a meaningful human act.

Further Reading and Sources

THIS BIBLIOGRAPHY IS CURATED FOR THE PROFESSIONAL READER: EXECUTIVES, DESIGNERS, ENGINEERS, AND POLICYMAKERS WHO WANT TO GO DEEPER INTO THE ARGUMENTS OF THIS BOOK. IT IS DELIBERATELY SHORT ON THEORY FOR ITS OWN SAKE AND LONG ON SOURCES THAT HAVE SHAPED HOW LEGACY INDUSTRIES ACTUALLY OPERATE. ENTRIES ARE GROUPED BY THEME, AND EACH CARRIES A NOTE ON WHAT IT IS AND WHY IT MATTERS. WHERE A SOURCE IS AVAILABLE ONLINE — MOST INDUSTRY REPORTS ARE — THE PUBLISHER OR ISSUING BODY IS NAMED SO THE READER CAN LOCATE THE CURRENT EDITION.

Industrial Design History and Theory

Donald A. Norman, *The Design of Everyday Things* (Basic Books, revised edition 2013). The indispensable starting point for anyone who wants to understand design thinking. Norman's account of affordances, signifiers, and the gap between how designers imagine users and how users actually behave is the foundation of human-centered design, and it applies to a control room or a mining truck cab as directly as to a door handle. Every designer in industry should have read it; most executives should too.

Donald A. Norman, *The Design of Everyday Things*' companion volume: *Emotional Design: Why We Love (or Hate) Everyday Things* (Basic Books, 2004). The bridge from usability to emotion, and the theoretical backbone of the argument that industrial products can and should be loved. Norman's three levels of processing — visceral, behavioral, reflective — explain why a farmer can love a tractor and why aesthetics is a competitive weapon in heavy industry.

Leander Kahney, *Jony Ive: The Genius Behind Apple's Greatest Products* (Portfolio, 2013). The most accessible account of how a designer rose to the center of a corporation and made design the company's strategy. Ive's career is the proof case for the argument that design leadership belongs at the board level, and his methods — obsessive prototyping, integration of hardware and software, a design studio with real authority — are directly transferable to industrial settings.

John Heskett, *Industrial Design* (Thames & Hudson, 1980) and *Design: A Very Short Introduction* (Oxford University Press, 2002). Heskett was the most lucid historian of industrial design's economic role. His central insight — that design is the interface between what things do and what they mean, and that its economic value is consistently underestimated — underpins the commercial arguments throughout this book.

Victor Papanek, Design for the Real World (Pantheon, 1971; reissued many times). The founding text of socially responsible design. Papanek's attack on design as a servant of consumerism, and his insistence that design must serve human need, anticipated the sustainability and dignity arguments that now dominate industrial practice. Dated in places, prophetic overall.

Adrian Forty, Objects of Desire: Design and Society Since 1750 (Thames & Hudson, 1986). A materialist history of design that traces how industrial objects carry ideology and identity. Forty's chapters on the design of office equipment and domestic machinery show that the emotional and symbolic life of industrial objects is not a modern marketing invention.

The Manufacturing Renaissance

Gene Kim, Kevin Behr, and George Spafford, The Phoenix Project (IT Revolution Press, 2013). The unlikely bestseller that framed manufacturing-style bottlenecks, constraints, and continuous improvement for the digital age. Told as a novel, it is the fastest way for a design or engineering professional to absorb the language of value streams, throughput, and the Theory of Constraints without reading a textbook.

Eliyahu M. Goldratt and Jeff Cox, The Goal (North River Press, 1984; 30th anniversary edition 2014). The original business novel and the source of the Theory of Constraints. Every argument about supply chain

fragility and resilience in this book traces back to Goldratt's point: optimize the whole system, not the parts, and measure against the goal, not against local efficiency.

Peter Marsh, *The New Industrial Revolution: Consumers, Globalization and the End of Mass Production* (Yale University Press, 2012). The best single-volume account of how manufacturing is changing, written by a longtime Financial Times manufacturing correspondent. Marsh's thesis — that we are moving from mass production toward more customized, geographically diverse, skill-intensive production — anticipated the reshoring and regionalization trends discussed in this book.

Rana Foroohar, *Makers and Takers: The Rise of Finance and the Fall of American Business* (Crown, 2016). A polemic, but a well-evidenced one, about how financialization hollowed out Western manufacturing. Essential context for understanding why the industrial policy reversal of the 2020s happened, and why design and engineering capability were allowed to decay in the first place.

Vaclav Smil, *Made in the USA: The Rise and Retreat of American Manufacturing* (MIT Press, 2013). The definitive quantitative account of American manufacturing's decline and its structural causes. Smil is the great antidote to both boosterism and doom: he shows what manufacturing actually contributes to an economy, and what its retreat actually cost.

Dan Gardner and Philip Zelikow, The New Industrial Age is preceded by the pandemic-era supply chain literature; the most useful single volume is Willy C. Shih's and Gary P. Pisano's Producing Prosperity: Why America Needs a Manufacturing Renaissance (Harvard Business Review Press, 2012). The academic case for why manufacturing capability — not just services, not just innovation — is essential to national prosperity. Shih and Pisano's concept of the "industrial commons" explains why offshoring erodes not just jobs but the ecosystem of skills and suppliers that makes innovation possible.

Industry and Strategy Reports

World Economic Forum, The Future of Jobs Report (biannual). The most cited global forecast of labor market change, with detailed sectoral data on the skills that will be in demand and the jobs that will be transformed. The manufacturing and engineering chapters are directly relevant to the workforce arguments in this book. Free to download from weforum.org.

McKinsey Global Institute, The Future of Work in Europe (2020) and Reshaping Supply Chains: Reimagining Global Production (2020). McKinsey's supply chain research — including the finding that companies could redesign global supply chains for resilience at modest net cost — is the empirical basis for the resilience chapter of this book. Also worth reading:

Manufacturing the Future: The Next Era of Global Growth and Innovation (2012), which remains the clearest statement of advanced manufacturing's economic stakes.

Boston Consulting Group, Resilient, Sustainable, and Digital: The New Manufacturing Operating Model (annual reports from 2021). BCG's annual surveys of manufacturing executives track the reshoring decision in real time, including the share of companies moving production closer to customers and the reasons they cite. The data on cost penalties versus resilience benefits is the best available.

World Economic Forum and McKinsey, The Global Lighthouse Network reports (annual). The WEF's "lighthouse factories" program documents the world's most advanced production sites — including the Siemens, BMW, and Schneider Electric plants cited in this book — with quantified results for productivity, sustainability, and workforce impact. The clearest public evidence that the factory of the future is not a concept but an operating reality.

Design Council (UK), The Design Economy reports (periodic). The most rigorous attempt to measure design's economic contribution at a national level, including the value of design to manufacturing sectors. The Design Council's finding that design-intensive businesses outperform their sectors is the empirical anchor for the argument that design is a competitive asset, not a cost.

European Commission, Industrial Strategy updates and the Critical Raw Materials Act and Net-Zero Industry Act documents (2020-2024). The primary sources for European industrial policy. The Commission's own impact assessments are the best available summary of Europe's dependencies and the policy logic of strategic autonomy.

US Department of Commerce and Semiconductor Industry Association, Winning the Future: A Blueprint for US Leadership in Semiconductor Manufacturing (2021), and the CHIPS and Science Act text and implementing guidance. The primary record of the American attempt to rebuild domestic semiconductor production, including the design and workforce conditions attached to federal funding.

Journals

Design Studies (Elsevier). The leading academic journal on design research, with a strong tradition of studying design in engineering and manufacturing contexts. Articles on design cognition, design methods, and the role of design in product development are directly relevant to the professional reader.

She Ji: The Journal of Design, Economics, and Innovation (Tongji University, open access). A newer journal that explicitly bridges design, economics, and innovation policy — the exact intersection this book

occupies. Its special issues on design-driven innovation and on design in China and the West are particularly useful.

Design Issues (MIT Press). The longest-running journal of design history and theory. Essential for the historical arguments about industrial design's evolution, and for the debates about design's social and political role.

Journal of Industrial and Production Engineering (Taylor & Francis). The engineering counterpart to the design journals: production systems, manufacturing technology, and human factors in manufacturing. Useful for the reader who wants the technical detail behind the strategic arguments.

International Journal of Human-Computer Studies and Applied Ergonomics (Elsevier).** The two journals most relevant to the design of industrial work: human factors, interface design, and the ergonomics of aging and assistive technology. The research base for the workforce arguments in this book lives here.

Harvard Business Review and MIT Sloan Management Review.** Not journals in the academic sense, but the most reliable source of practitioner-grade analysis on servitization, digital transformation, and manufacturing strategy. The HBR articles by Shih, Pisano, and Porter on manufacturing competitiveness remain the standard citations.

Organizations and Institutions

Design Council (UK). The UK's national design authority and the source of the most credible design-economy data in Europe. Its design methods framework (the Double Diamond) remains the most widely used process model in professional practice. designcouncil.org.uk

Industrial Designers Society of America (IDSA). The professional body for industrial designers in the United States, and the organizer of the International Design Excellence Awards (IDEA), one of the three major global design competitions alongside the Red Dot and iF awards. idsa.org

International Council of Design (ico-D). The global federation of design organizations, useful for its international design education standards and its surveys of design practice across economies. theicod.org

Design Management Institute (DMI). The professional home for design managers and design leaders, and the publisher of the DMI Design Value Index — the longitudinal study showing that design-led public companies have outperformed the S&P 500. The index is the single most cited statistic in the business case for design. dmi.org

Fraunhofer-Gesellschaft. Germany's applied research network, whose institutes (especially IAO for work organization and IFF for factory automation) are the

world's most systematic link between research, design, and industrial practice. The model most countries are trying to copy. fraunhofer.de

The High Value Manufacturing Catapult (UK). The network of UK research centers connecting universities to manufacturers, with capabilities spanning design, materials, and production. The best British example of policy infrastructure for industrial design capability. hvm.catapult.org.uk

The World Economic Forum's Platform for Shaping the Future of Advanced Manufacturing and Production. The convening body behind the lighthouse factory network and the most influential global forum for manufacturing strategy. Its reports and communities are open to practitioners and are the best way to track the frontier of industrial practice. weforum.org

Key Standards

ISO 9241 — Ergonomics of Human-System Interaction. The foundational standard family for ergonomics and usability, covering workstation design, display ergonomics, and (in its 210 series) human-centered design processes. The professional baseline for every design-for-the-worker decision in this book. The related ISO 6385 (principles of ergonomics in work system design) extends it to whole work systems.

ISO 13407 (superseded by ISO 9241-210) — Human-Centred Design for Interactive Systems. The process standard for user-centered design: understanding context, specifying requirements, producing design solutions, and evaluating against them. The standard's discipline is the closest thing the profession has to a universal design process.

ISO 9001 — Quality Management Systems. The management standard that every industrial supplier must meet, and the gateway through which design quality is audited in most supply chains. Designers who understand how design decisions are judged under ISO 9001 (design and development clauses) will navigate industrial organizations far more effectively.

ISO 14001 — Environmental Management Systems and the emerging **ISO standards for circular economy (ISO 59000 series).** The environmental management standards that increasingly govern industrial design, and the new circular economy framework that will shape design for disassembly, reuse, and recycling over the coming decade.

IEC 61508 / ISO 26262 — Functional Safety (general and automotive). The safety standards that govern any design involving electronics, software, and risk — which is to say, virtually all modern industrial products. Understanding functional safety is now a core competency for industrial designers working with autonomy and automation.

ANSI/ISA-101 — Human-Machine Interfaces for Process Automation. The emerging US standard for industrial HMIs, codifying the difference between good and bad operator interfaces in control rooms and on equipment. The closest thing to a design specification for industrial software, and a direct expression of the arguments about the digital layer in this book.

The Machinery Directive (2006/42/EC) and the EU AI Act (2024). The regulatory frameworks governing machine design in Europe, now extended to artificial intelligence. For any designer working with automated or intelligent machinery, these are the legal boundaries of the design space — and the AI Act's human-oversight requirements give regulatory force to this book's argument that data and automation must serve human judgment, not replace it.

Toolkit Summary

- Start with the canon: Norman for design thinking, Kahney for design leadership, Heskett and Forty for industrial design's history and economic role.
- For the manufacturing renaissance, read the business novels (The Phoenix Project, The Goal) for the mindset and Marsh, Smil, and Shih and Pisano for the evidence.

- Track the present with the WEF Future of Jobs Report, McKinsey's supply chain research, BCG's reshoring surveys, and the Global Lighthouse Network.
- Follow the journals that matter: Design Studies, She Ji, Design Issues, and Applied Ergonomics — and the practitioner analysis in Harvard Business Review.
- Join or follow the institutions: Design Council UK, IDSA, DMI (home of the Design Value Index), Fraunhofer, and the High Value Manufacturing Catapult.
- Know the standards: ISO 9241 for ergonomics, ISO 9241-210 for human-centered process, ISO 9001 and 14001 for management systems, IEC 61508 and ISA-101 for safety and interfaces, and the Machinery Directive and EU AI Act for the legal boundary of design.

About the Author

Marcus Hale is a design strategist and writer with twenty years of experience across manufacturing, heavy industry, and product development. He has worked with legacy companies in aerospace, automotive, and industrial machinery as they navigate digital transformation.

His writing focuses on the intersection of industrial design, technology, and the human systems that keep industry alive. He believes the next industrial age will be defined not by what we build, but by how thoughtfully we redesign what already exists.

Marcus lives in the North of England and is a passionate advocate for the revival of Western manufacturing.

Also by Marcus Hale

The Next Industrial Age: The Future of Industrial
Design in Western Legacy Industries

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